CHAPTER

18

Dentin Hybridization: Bonding Agents

One major problem in restorative dentistry is the lack of proper union between restorative material and the tooth surface. The gap at the tooth–restoration interface may create problems, such as sensitivity and recurrent caries, etc; subsequently, failure of the restoration. The continuous inventions have led to the development of various techniques and modalities, which help in adhesion/bonding. Bonding or adhesion can be physical, mechanical or chemical; such restorations are known as 'bonded restorations' or 'adhesive restorations'. However, the researchers have not been able to minimize the restoration–tooth interface, because of many inherent weaknesses of restorative materials like setting/polymerization expansion/contraction, different coefficients of thermal expansion, modulus of elasticity, etc.

Bonding improves retention and stabilization of a restoration without excessive removal of sound tooth structure. The need for providing auxiliary retentive aids is eliminated. Adhesive restorations are better able to transmit and distribute functional stresses, thereby reinforcing weakened tooth tissue. Bonding also facilitates repair and replacement of deteriorated fillings with little or no additional removal of tooth structure.

Adhesive techniques have greatly expanded the horizon of esthetic dentistry. Correction of shapes, positions, dimensions and shades of teeth is convenient with the adhesive restorative materials. Repair of fractured teeth can be carried out using the same fractured fragments thereby maintaining original esthetics. Bonding reduces the extent and amount of microleakage. Preventing the ingress of oral fluids and bacteria along the cavity–restoration interface reduces most clinical problems, such as post-operative sensitivity, marginal staining and recurrent caries.

Adhesive agents are also useful in other clinical situations, viz. luting porcelain, composite or metal inlays, onlays, crowns, veneers, fixed partial dentures, resin-bonded prosthesis, etc. These agents can also be employed for treating dentin hypersensitivity, as pit and fissure sealants, in direct pulp capping procedures and beneath amalgam restorations as insulators.

Before studying the bonding agents, let us discuss the basic principles involved in bonding phenomenon.

COMMON TERMS: DEFINITIONS

Adhesion: Adhesion is the force or the intermolecular attraction that exists between molecules of two unlike substances when placed in intimate contact with each other.

Adhesion in dentistry is of three types, involving the following mechanisms:

1. *Chemical adhesion* relies on primary valence forces, such as covalent, ionic or metallic bonds
2. *Physical adhesion* relies on secondary valence forces, such as van der Waals force, hydrogen bond, etc.
3. *Mechanical adhesion* relies on penetration of one material into a different material at microscopic level, such as formation of hybridized dentin; resin polymers get entangled with collagen fibrils of dentin.

Adhesive/adherend: The substance added to produce adhesion is known as the '*adhesive*' and the material to which applied, is known as the '*adherend*' (Fig. 18.1). An interface is present wherever adhesion exists. Adhesion can be seen between any two phases, e.g. solid, liquid or gas with the exception of two gases, where an interface is not present. Most commonly, a solid is the adherend and liquid is the adhesive.

Interface: The junction between two dissimilar materials (generally considered as two-dimensional; say junction between metal and resin).



Fig. 18.1: Adhesive and the adherend

Interphase: The junction between two dissimilar materials in three dimensions (e.g. Hybrid layer is interphase among composite resin, adhesive and the tooth substrate).

Hybridization: The process of creating a hybrid layer in dental hard tissues.

Hybrid layer: A layer formed in dental hard tissues by demineralization, followed by infiltration of monomer and subsequently polymerization.

Resin tag: The extension of adhesive resin into demineralized (open dentinal tubules) dentin.

Hybridized tag: In case, the resin tags get attached to the tubule walls (peritubular dentinal matrix of top 2–4 μm is removed, subsequently monomer diffuse into the surrounding dentinal matrix).

Bioaffinity: Substances having capability of binding to biological materials are categorized as having bioaffinity.

FACTORS AFFECTING ADHESION

Most adhesives rely on van der Waals forces plus the mechanical adhesion, if available. If the surface is rough and porous, the mechanical adhesion will predominate. If the surfaces are smooth and polished, van der Waals forces are effective. Although theoretically, an adhesive bond that sets up by primary chemical bonding should reveal higher adhesive strength than one that relies on van der Waals bonds; but in practice, the chemical bonds are susceptible to attack by water, and often prove weaker than the van der Waals bonds.

The phenomenon of adhesion is dependent upon certain factors. The three factors, surface energy, wetting and contact angle are important determinants of adhesion. These factors control adhesion and are also closely interrelated.

a. ***Surface energy:*** The energy of a solid on the outer surface is comparatively higher than its interior. Inside the crystal, each atom is equally surrounded by atoms on all sides and the interatomic distances are equal, hence the energy is minimal; whereas, towards the periphery, the atoms are not equally distributed. The surface atoms get strongly attracted to each other resulting in surface tension. Because of this energy, substances are attracted to the substrate surface; for example, silver, gold and platinum attract oxygen molecules to their surface. With gold, oxygen adheres by secondary forces; whereas with silver, chemical forces exist forming silver oxide. The adhesive can attach to the adherend either by physical or chemical force. Initially when two substances are far apart, only physical forces exist between them but as the distance diminishes, chemical forces start becoming effective. In practice, when two attracting molecules are separated by a distance that exceeds their own approximate diameter, the distance usually being 3.0–4.0 nm, they no longer show any appreciable attraction. Hard solids have specific free surface energy values between 500 and 5000 ergs/cm^2. Harder the surface, higher is the surface energy and higher the adhesive properties.

b. ***Wetting:*** When two glass slides (may be polished) placed in close contact with each other, do not exhibit any attachment. This owes to the presence of microirregularities on the slides and is responsible for the two surfaces contacting only at their 'high spots' or 'hills'. The total surface area in contact is decreased, and hence no perceptible adhesion takes place; whereas, if a film of water is placed between them, it becomes increasingly difficult to separate the two. For producing such type of adhesion, the liquid must easily flow over the entire surface and adhere to the solid. This characteristic is referred to as wetting. The wetting ability of a liquid adhesive depends upon the surface energy and cleanliness of the adherend. Higher the surface energy, greater would be the wetting capability. Metals generally have a high surface energy. On the other hand, teflon (polytetrafluoroethylene) synthetic resin has a comparatively very low surface energy. Contamination of the substrate surface also reduces wetting by the adhesive.

c. ***Contact angle:*** Contact angle is a measure of wettability and is the angle formed by the adhesive with the adherend at the interface (Fig. 18.2A to C). Smaller the contact angle, greater is the wettability of the adhesive. A low viscosity of the adhesive is imperative for better flow and bond formation.

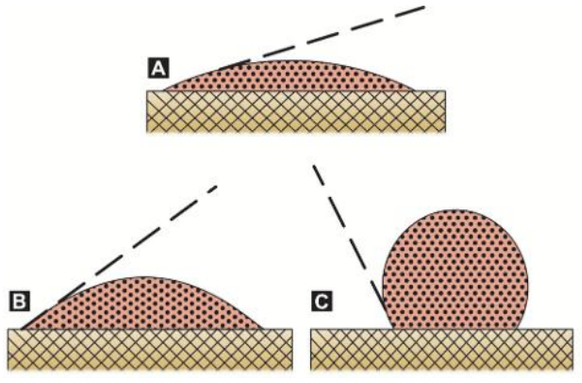


Fig. 18.2: Degree of contact angle influences the wetting of surface; **(A)** When contact angle is small, wettability of the adhesive is better; **(B** and **C)** As the contact angle is increased, wettability is decreased

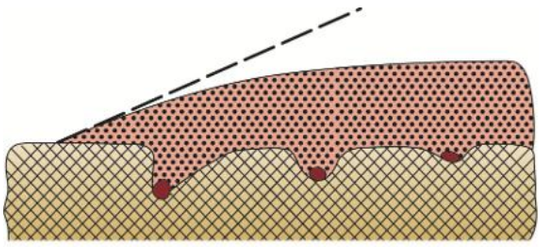


Fig. 18.3: Air incorporated in the irregularities at the interface despite low contact angle of the adhesive. Such areas are points of stress concentration and promote adhesive failure

However, microirregularities and crevices remain a limiting factor in close bond formation, because air may be entrapped at the base and serve as discontinuities in the adhesive joint (Fig. 18.3). Under continual thermal and mechanical loading, stress concentration occurs around these sites; subsequently, initiating break adjacent to the void.

Many a times, there is only mechanical union between two surfaces (attachment of two substances by mechanical means rather than molecular bonding). At a gross level, examples include screws, bolts or undercuts.

Retention of cast gold restoration using luting cement is also dependent on the amount of mechanical union between the two surfaces.

TOOTH AS A SUBSTRATE FOR BONDING

The tooth surface, used as a substrate for bonding, comprises of two components: Enamel and dentin. Though bonding to enamel is effectively achieved in clinical practice, similar level of adhesion to dentin poses difficulties. Significant contrast in composition and structure of enamel and dentin greatly influence the prospects of bonding. Both organic and inorganic components vary in both the tissues (Table 18.1).

Within the same tissue, nature of the substrate presented for bonding as may vary with the location, e.g. enamel etched on its rod ends provide better area for bonding as compared to enamel etched on the side of the prisms. Fluoridated enamel has a low surface energy and is more difficult to bond as compared to the nonfluoridated enamel. Dentin in its superficial portion allows greater bond strengths than in the deeper portions. This might be because the superficial layer has more of solid dentin and less moisture contamination as compared to deeper portions. Sclerotic areas present difficulties in bonding, as compared to natural tubular dentin.

The presence of moisture in dentin is the major hurdle in achieving bonding. The inherent wetness owes to the outward flow of the dentinal fluid under positive pulpal pressure. Occasionally, moisture may be seen on the surface of dentin due to external factors, like accidental contamination by saliva and/or air humidity. In order to obtain a stable and long-lasting bond to wet dentin, it is essential that the adhesive gets displaced in the dentin fluid. Furthermore, such a bond must be maintained in a continuously aqueous environment. Dentin permeability is more towards the pulp horns and near the pulp, because of increase in tubule density towards pulp horn and tubule diameter near pulp.

Permeability reduces with the presence of intratubular deposits, tubule irregularities, etc.

An important factor that influences the bonding to tooth surface is the presence of smear layer on cut dentin. The smear unit (smear layer + smear plugs) may serve as a contaminant and prevent adequate penetration of the adhesive into the underlying substrate; a few authors believe that it acts as a protective barrier to the pulp by minimizing leakage of bacteria and other by-products.

Requisite of an Ideal Bonding Agent

- Biocompatible
- Bond effectively to both enamel and dentin
- Sufficiently strong to resist failure as a result of masticatory forces
- Mechanical properties close to tooth structure
- Resistant to degradation in oral environment
- Easy to use (manipulation)

BONDING TO ENAMEL

Conditioning and Etching Enamel

Enamel surface generally has a low surface energy, inert, reactive and under normal circumstances maintains homeostasis with the surroundings. Untouched enamel is covered by a predominantly organic layer, comprised of adsorbed proteins most of which originate from the saliva. This organic biofilm is referred to as the *'pellicle'*. Enamel when

Table 18.1: Composition of enamel and dentin

Component	*Enamel*	*Dentin*
Inorganic (hydroxyapatite)	94–95%	60–70%
Organic (collagen)	4–5%	20–25%
Water	1.0–1.5%	10–15%
Calcium phosphate ratio	1:64	1:56

ground, is also covered by a protein-rich film called the '*smear layer*'. Both pellicle and smear layer are considered as surface contaminants, which differ from their subsurface in composition and structural arrangement. In stagnant sites, these protein biofilms on enamel may attract micro-organisms forming plaque. It is evident that untreated enamel presents a less than ideal surface for establishing adequate bonds. The objective is to remove the contaminants and subsequently raise the energy and reactivity of the enamel surfaces. The procedure that leads to this removal of organic layer, making the enamel surface more reactive is known as '*conditioning*'. The procedure that leads to demineralization of superficial calcium ions is known as '*etching*'.

A thorough dental prophylaxis for removing materia alba, plaque and other accretions is an important component of the conditioning/etching regime. It has been observed that prophylaxis alone can double the bond strength. The prophylaxis pastes devoid of oils, flavoring agents and fluorides are recommended for this purpose. Rubber cups are preferred as they are less likely to damage gingival tissue or abrade enamel. After cleaning, the enamel is thoroughly washed with water, the treatment site dried and carefully isolated from oral fluids, like saliva and gingival crevicular fluid. Ever since Buonocore (1955) used phosphoric acid for conditioning/etching enamel, researchers have tried various concentration of phosphoric acid in the form of liquid or gel, alternative acids and varying etching times. Less concentration of acid for smaller period of time is used for conditioning, while the higher concentration of the same acid for longer period is used for etching. Following acid application for stipulated period, the area is thoroughly washed for 10–15 seconds to remove the acid, its reaction products and mineral hydroxyapatite. *In case gels are used, the washing time is doubled, so as to flush away its cellulose vehicle, which otherwise might serve as a contaminant*. An effectively etched surface on drying gives a matt white or frosted appearance. Since the etched surface at this stage has a very high surface energy, a contaminant can readily adhere to the surface. Even a minor exposure to saliva, blood or oil can ruin the potential for resin tag formation and bonding. All measures should be taken to prevent any contamination. If accidental contamination occurs, the procedure should be repeated.

Etching Pattern

The patterns of etching in enamel are:

Type I: Most common etching pattern, involves the preferential removal of enamel prism cores with prism peripheries remaining relatively intact.

Acid conditioning/etching: Key effects

- Removes residual pellicle to expose the inorganic crystallite component of enamel.
- Removes approximately 10 µm of surface enamel.
- Creates a porous layer, the depth of the pores range from 5.0–50.0 µm. When a low viscosity resin is subsequently applied, it flows into the microporosities and polymerizes to form resin tags.
- Increases the wettability and surface area of the enamel substrate.
- Raises the surface energy of enamel with creation of reactive polar sites.

Type II: Etching pattern is the reverse of type I, i.e. the peripheries are removed leaving the cores intact.

Type III: Etching pattern is less distinct, and includes areas resembling type I and type II patterns as well as regions in which the etching pattern appears unrelated to prism morphology. This type of etching is generally associated with the presence of prismless enamel.

Type IV: Pattern displays only a random distribution of depressions with no preferential destruction of either cores or peripheries. These pitted areas occasionally occurred in little patches over the enamel surface. This type of etching pattern is commonly seen in cervical areas and rarely on occlusal.

Type V: Pattern shows no evidence of prism outline. The regions of enamel are flat and smooth and lack microirregularities for penetration of resin. Such type of pattern is seen in high fluoride areas.

Resin tags, which form in the enamel rod peripheries, (i.e. between enamel prisms) are known as '*macrotags*'. A much finer network of thousands of small tags form across the end of each rod where individual hydroxyapatite crystals have been dissolved leaving crypts outlined by residual organic matter. These fine tags are called '*microtags*'. Both macrotags and microtags are the basis of micro-mechanical bonding to enamel (Figs 18.4 and 18.5). Microtags are essentially more important because of their larger number and also they offer greater surface area of contact. It is established that when enamel prisms are exposed perpendicularly, the bond strength is reduced to 50% as compared with the parallel exposure of the prisms.

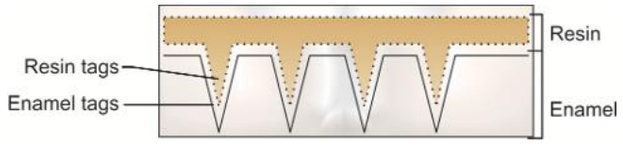


Fig. 18.4: Resin tags and enamel tags aid in micromechanical bonding

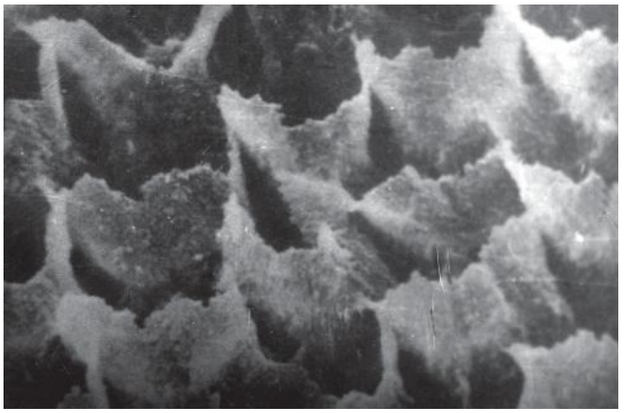


Fig. 18.5: Magnified view of enamel tags in which resin tags will fit

Phosphoric Acid as an Etchant

Thirty-seven percent phosphoric acid is the most commonly used acid for etching. Higher concentrations (more than 50%) may lead to formation of a layer of monocalcium phosphate monohydrate on etched surface, which prevents any further dissolution; whereas, concentration less than 30% forms a precipitate of dicalcium phosphate dihydrate that cannot be easily removed. Higher concentration of acid does produce deeper tags; however, being brittle give way, leaving 10–25 µm depth tags as obtained with the lower concentration. It is accepted that 30–40% concentration of the acid is the most effective.

Acid etchants are available in liquid and gel form (Fig. 18.6A, B). Gel etchants are preferred because of the ease and control of placement. The etching pattern with gel and liquid remains the same. Gels are often made by adding colloidal silica or polymer beads to the acid. It is observed that phosphoric acid made into gel with silica had higher pH than phosphoric acid gel with polymer. These additives change the pH of these acids. They are applied to the surface either with a brush or dispensed through a syringe onto the enamel.



Fig. 18.6: Acid etchant: **(A)** Liquid form; **(B)** Gel form

The time for etching with 30 to 40% phosphoric acid has been recommended as 15 seconds. Studies with Scanning Electron Microscopy have confirmed that 15 seconds etching provides the same surface roughness as 60 seconds. Also, similar shear bond strengths and marginal leakage values for 15 and 60 seconds have been observed. However, enamel which is acid resistant because of a high fluoride content can be etched for 60 seconds. *Primary teeth also require longer etching time, since the enamel is more aprismatic than that of permanent teeth.*

Alternative acids have been tried as enamel etchants; mainly, pyruvic acid and sulfuric acid. 2.0% sulfuric acid used for 30 seconds is as effective as phosphoric acid; whereas, higher sulfuric acid concentrations produce heavy crystal deposits, which interfere with bonding and cannot be washed away easily. With the present concept of total etch techniques, acids such as 10% phosphoric acid, 10% maleic acid, 2.5% nitric acid, citric acid, oxalic acid, etc. are used to etch both enamel and dentin simultaneously. However, weaker acids provide significantly lower shear bond strengths. Cupric acid added to 10% phosphoric acid may enhance the bond strength of 4-META based resins (cupric acid cross-link collagenous proteins, stabilizing interfibrillar spaces that are necessary for permeability of monomer during bonding).

Alternative Etchants

Though acid conditioning remains the most popular method of bonding, other methods have also been tried, viz:

i. Calcium sulfate crystals were deposited on the surface of enamel by reacting solution of polyacrylic acid and potassium sulfate. It was hypothesized that these crystals trapped resin and retain the same mechanically. However, subsequent studies showed that the etched surface under these crystals interfere with bonding.

ii. Lasers are being used for enamel/dentin etching. Laser etching is a process of continuous vaporization and microexplosions due to vaporization of water trapped within the hydroxyapatite matrix. In general, more material is removed by the

microexplosion of entrapped water than by direct vaporization of the hydroxyapatite crystals. The amount of surface roughening is dependent upon the type and wavelength of the laser. CO_2 and Nd:YAG are the most effective lasers for etching. It is observed that changes in surface morphology and bond strength after laser etching are similar to acid etching.

SEM images of enamel surface etched at varying time periods are shown in Fig. 18.7A to D.

Applying Bonding Agent

The bonding agents (low viscosity monomers) are flown over the etched enamel. It is believed that when applied to clean, dried and conditioned enamel, the bonding agent, because of its low viscosity, rapidly wets and penetrates into the microspaces forming resin tags. The resin also forms an extremely close physical relationship with the enamel crystallites and often encapsulates them to form a composite zone in the outer enamel. The bonding agent may be chemically cured or light cured. Subsequent placement of restorative resin becomes continuous with the resin-bonding agent. Bond strength to acid conditioned enamel using early adhesive agents have been reported to be in the range of 16–21 MPa. Over the period, bond strength have been greatly improved and may be attributed to better wetting agents, improved resin chemistry that allow complete polymerization of the adhesive and better water stability at the interface. Enamel bonding agents have been replaced by the same systems that are used on dentin as they facilitate simultaneous bonding to both enamel and dentin.

BONDING TO DENTIN

Dentin bonding is comparatively difficult because of the fact that dentin is a complex tissue and contains plenty of fluids. It consists of 50 vol% of inorganic content, 30 vol% of organic material, and 20 vol% of fluid. The high protein content is responsible for the low surface energy of dentin (44.8 dynes/cm^2), which differentiates it from enamel. Wetting of such a low energy surface is difficult vis-à-vis the adhesion. Further, water is an essential component of natural dentin and surface drying is difficult to achieve.

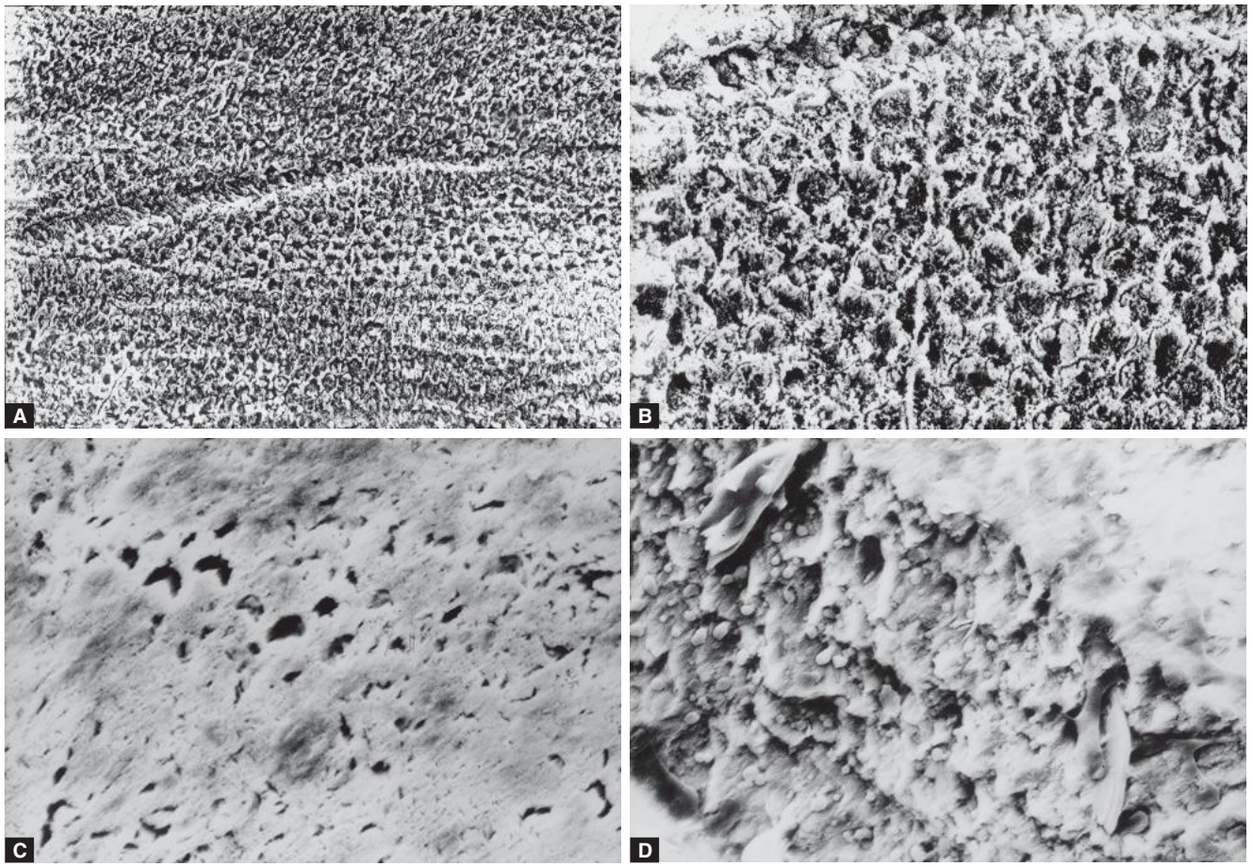


Fig. 18.7: SEM images of etched enamel surface: **(A)** Enamel surface etched for 15 seconds (surface view); **(B)** Enamel surface etched for 15 seconds (oblique section); **(C)** Enamel surface etched for 60 seconds (surface view); **(D)** Enamel surface etched for 60 seconds (oblique section)

Rewetting the surface, (may be subclinical), occurs instantaneously. The bonding is difficult in aqueous environment, since water competes effectively for the adhesion sites. Because of the vast structural and functional variations between the dentin and enamel, bonding to each is not similar.

Dentin is a living tissue and is directly connected to the pulp through fluid-filled channels that contain odontoblastic processes, the body of which lie in the pulp. The pulpal compatibility is very important when dentin bonding is concerned. It has been established that 37% phosphoric acid when used to etch both enamel and dentin, did not lead to pulpal damage.

Dentin Hybridization

The characteristics of successful bonding of a resin composite to dentin include micromechanical attachment between resin and demineralized, primed layer of intertubular dentin. This complex, referred to as the *hybrid layer*, is best achieved by acidic conditioning agents, which remove the smear layer, produce surface demineralization to a depth of 3.0–6.0 μm and expose the dentinal collagen framework. In some cases, the smear layer is not removed; but dissolved or modified to include it within the bonding process. Application of hydrophilic resin primers facilitates subsequent penetration of low viscosity adhesive resins into the microporous collagen scaffold and dentinal tubules. Polymerization of this infiltrated resin stabilizes the surface hybrid layer. Meticulous application of these steps leads to a gap-free bond.

The deep dentinal layer (dentin near the pulp) is composed mainly of large dentinal tubules with less intertubular dentin; whereas, superficial dentin has a few dentinal tubules, mainly composed of intertubular dentin.

Hybrid layer formation in intertubular dentin contributes to retention of resin. (The resin retention is proportional to the amount of available intertubular dentin.) Because the intertubular dentin area varies, the contribution of resin tag/hybrid layer to bond strength also varies.

The integration of resin tags into the hybrid layer is important to produce a resin seal of the dentinal tubules, contributing to resin retention.

Mechanism

Titley et al (1995) postulated three mechanisms for formation of resin tags in dentin. Acid etching produces funnel-shaped dentinal tubules on top and exposes the basement membrane-like sheath, called 'lamina limitans'.

i. If lamina limitans remain potent (no water, fluid or odontoblastic processes), then primers penetrate the tubules to a greater depth.
ii. If lamina limitans remain open (tubules filled with water, fluid or odontoblastic processes), the resin tags incorporate the sheath (inner zone may remain unfilled).
iii. If lamina limitans collapse and occlude the tubules, the resin penetration is limited to superficial zone only.

The resin tags may not always be funnel shaped. If the peritubular dentin matrix is not lost, the resin tags would have parallel walls (not funnel shaped). In case peritubular dentin matrix is removed with etching/conditioning (usual depth is 1.0–5.0 μm), the resin tags would be of funnel shaped.

Role of Smear Layer

Smear layer is a product of dental instrumentation, which covers the normal structural components of dentin by 1–2 μm and penetrates several micrometers (1–5 μm) into the tubules to form smear plugs. It has two phases, a solid phase of cutting debris—primarily denatured collagen and mineral, and a liquid phase of tortuous fluid-filled channels around the cut debris. Smear layer may be a deterrent to the bonding process, since it may serve as a barrier to the penetration of resin to the underlying dentin substrate (Fig. 18.8). Bacteria entrapped in the smear layer can survive and multiply beneath the restoration. A few authors, however, believe it to be a protective barrier against bacterial and toxin penetration.

Reasons for retaining smear layer are:

- Retention of smear layer lowers dentin permeability.
- Prevents decrease in bond strength seen with some bonding systems as deeper dentin is prepared.
- Greatly lowers the effect of pulpal pressure on bond strength.

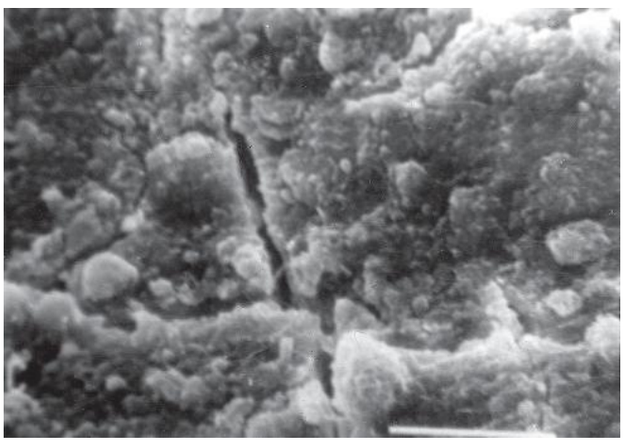


Fig. 18.8: Presence of smear layer on cut tooth surface

The present dentin adhesives involve modification of the smear layer as it is believed to greatly improve the bond strength to dentin. The removal of smear layer and demineralization of dentin matrix may facilitate bonding through a number of mechanisms like:

- The exposed collagen provides reactive groups that can chemically interact with primers.
- Amino groups may act as a catalyst to polymerization reactions.
- Exposed collagen promotes micromechanical bonding to resin by providing a framework.

It is quite probable that all three mechanisms operate simultaneously to improve bonding.

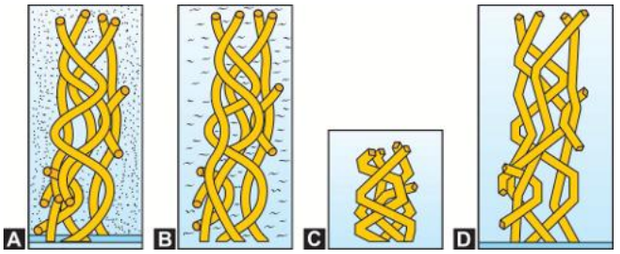


Fig. 18.9: Schematic representation of dentin matrix: **(A)** Mineralized dentin matrix; **(B)** Demineralized dentin matrix filled with water, i.e. in a plasticized state; **(C)** Collapsed, stiffened, air-dried demineralized dentin matrix; **(D)** Demineralized dentin matrix stiffened by organic solvent prior to air drying

Conditioning of Dentin

Conditioning of dentin is defined as "*an alteration of the dentin surface including the smear layer with the objective of producing a substrate capable of micromechanical and possibly chemical bonding to a dentin adhesive*". The principal effects of conditioning on dentin can be physical or chemical. Physical effects are the alteration in the thickness and morphology of the smear layer and also the dentinal tubules; whereas, chemical effects are modifications of the fraction of organic matter and decalcification of the inorganic portion.

Conditioning can be performed by chemical, thermal (laser) or mechanical means.

A. Chemical Conditioning

Various acids and calcium chelators, which rely on removing the smear layer, are used as chemical conditioners.

i. ***Acid conditioners:*** Acid conditioners not only remove the smear layer, but also simultaneously demineralize superficial dentin (3–6 μm). On intertubular dentin, the collagen fibers are often covered by an amorphous layer, which is of variable thickness and has microporosities. This 'collagen smear layer' may reduce the permeability of underlying dentin to resin monomers, and is insoluble in acids. At the tubule orifices, peritubular dentin is often completely dissolved to form a funnel shape structure and expose collagen fibrils, which are often additional retentive sites at the tubule wall (Fig. 18.9 A to D). After conditioning, maintenance of a moist dentin surface is recommended, following the wet bonding technique to prevent collapse of unsupported collagen and promote wetting and infiltration of subsequently applied resin. 37% phosphoric acid is routinely used. 10% phosphoric acid appears to provide better bond strength than the higher concentrations. Dentin bonding agents (e.g. Tenure, Mirage bond) that use nitric acid conditioners, are highly adhesive and provide good tubule seals. 10% citric acid combined with 3.0% ferric chloride is used as a smear layer remover and etchant. Another combination etchant is 10% citric acid and 20% calcium chloride (e.g. Clearfil Liner Bond). Compared to the phosphoric acid etching (Fig. 18.10A), this combination etchant decalcifies

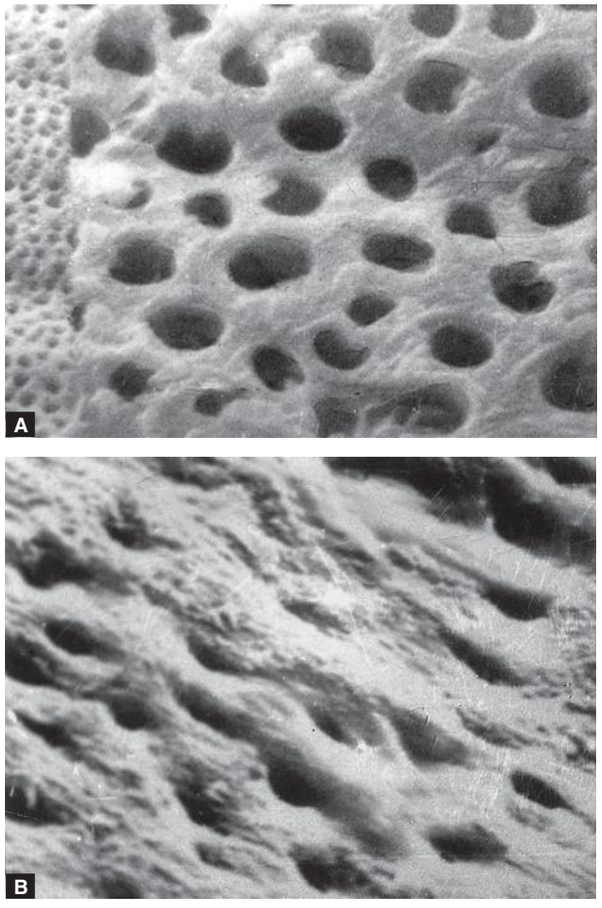


Fig. 18.10: SEM images of etched dentin surface: **(A)** Dentin surface after phosphoric acid etching; **(B)** Dentin surface after conditioning with a combination etchant of 10% citric acid and 20% calcium chloride

dentin to a lesser depth, and tubules do not open into a funnel shape (Fig. 18.10B). Maleic acid (e.g. Scotch bond) results in removal of the smear layer, but not the smear plugs.

ii. ***Chelators:*** Contrary to the strong acid etchants, chelators remove the smear layer without decalcification or significant physical changes on the underlying dentin substrate. Also, no funnel-shaped changes are evident; instead, a chelate is formed. A chelate refers to *a compound with a central metal ion surrounded by covalently bonded atoms, ions or molecules called ligands, which possess additional bonds for chemical reaction*. The best chelator conditioner is ethylenediaminetetracetic acid (EDTA, pH: 7–4), developed for use in Gluma system. When used for 30 seconds, the smear plugs were not fully removed with this conditioner.

Certain bonding systems do not remove the smear layer, rather utilize it as a legitimate bonding substrate. These systems utilize self-etching/self-priming agents, which are acidic monomer and can enlarge the channels and also remove mineral deposits from within the tubules; thereby promoting resin permeation. With this type of bonding, there are less chances of incomplete resin penetration because both etching and priming occur simultaneously. Since the self-etching/self-priming solution is not rinsed from the surface, the demineralized smear layer is incorporated into the bonding interface.

B. *Thermal (Laser) Conditioning*

Lasers are being used to condition dentin. The commonly used lasers are CO_2 and Nd:YAG.

It is speculated that lasers cause recrystallization of dentin, resulting in a fungi form appearance, that contributes to increased microretention or possible chemical adhesion of a restorative material to the tooth structure. The carbonized black spot that results after lasing, is easily washed off with water.

C. *Mechanical Conditioning*

Microabrasion is used to mechanically condition the dentin.

Priming

Priming is the second step in bonding procedure. The primers are the agents that contain monomers with hydrophilic properties, which have an affinity for the exposed collagen fibrils, and hydrophobic properties for copolymerization with adhesive resin. These monomers, usually HEMA and 4-META, are dissolved in solutions of acetone or ethanol, which because of their volatile characteristics can displace water from the dentinal surface and the moist collagen network. The primer promotes resin diffusion into the moist, demineralized dentin with the aim of achieving complete resin penetration. Microscopic examination has, however, shown deficiencies in the spread of primer at three levels: (i) Incomplete surface coverage, (ii) incomplete interfibrillar penetration, and (iii) incomplete penetration to the full depth of demineralized dentin. In order to achieve the best bond strength, it is essential that the primer spreads uniformly over the surface.

To improve spread and diffusion of primer, it can be applied in multiple coats. When a second coat of primer is applied, the shear bond strength improves significantly; however, there is no further increase up to five additional applications. Second consideration for improving penetration is the condition of the dentinal surface. As described earlier, water is an essential component needed to keep the collagen fibrils suspended; hence, creating space for subsequent penetration of primer and adhesive resin. Excessive drying of the conditioned dentin collapses the collagen network and the primer is prohibited from entry. On the other hand, an overwet surface results in separation of the primer components leading to emulsion polymerization of the adhesive resin. An ideal dentin surface for bonding is visibly moist, without any excessive water. This can be achieved clinically by blot drying with a damp cotton pledget. Third, short conditioning time (not more than 15 seconds) is considered adequate for allowing complete penetration of primer and resin. It is speculated that more stable bonds are formed, if primer and resin permeate dentin less deeply and more uniformly, rather than spotty and more deeply.

DENTIN BONDING AGENTS

Bonding agent is an unfilled or semifilled resin, which has a lower viscosity to permit easy flow and penetration. The major role of adhesive resin application is the formation and stabilization of hybrid layer, production of resin tags in the unplugged dentinal tubules and at the same time effectively sealing them to reduce the chances of increased permeability; subsequently, the pulpal irritation. When the adhesive resin is applied, a part of it penetrates into the microporous collagen scaffold of the intertubular dentin (the intertubular penetration). Here it polymerizes and copolymerizes with the primer to form an intermingled layer of collagen, termed as the *'resin reinforced layer'*, *'hybrid layer'* or *'resin interdiffusion zone'*. The rest of the adhesive resin enters into the dentinal tubules to a limited depth of 10–20 µm forming resin tags, (the intratubular penetration). Within the dentinal tubules, these tags are attached

only at this point to the peritubular dentin and any further extension of resin tags into the dentinal tubules is passive. Tags do not appear to contribute to the attachment mechanism; should be noted that prolonged conditioning is not recommended. With the present bonding systems, resins penetrate and hybridize the walls of lateral tubule, branches forming submicron resin tags; the phenomenon is called *lateral tubule hybridization*. Both intratubular and intertubular permeabilities are critical to achieve optimal bond strengths and sealing of dentinal surfaces. At times, the resin may not completely infiltrate into the collagen meshwork and such microspaces remain open to microleakage referred to as '*Nanoleakage*'. At two common sites of failure (top and bottom of hybrid layer), the resin tags keep the dentinal tubules sealed and the pulp protected. The ability of resins to penetrate both dentinal tubules and intertubular dentin is dependent upon several variables. In superficial portions of dentin, where tubules are fewer as compared to deep dentin, intertubular penetration accounts for the major portion of the bond strength; whereas, the reverse is true for deep dentin, where intratubular penetration contributes to most of the retention and sealing. Difficulties encountered in the infiltration of resins into tubules are mainly because of the presence of dentinal fluid and minerals in sclerotic dentin. Resin infiltration into intertubular dentin is primarily dependent upon the porosity between collagen fibrils, which further depends upon the amount of moisture in this layer.

Adhesive resins can either be chemically cured or light cured. For light-cured resins, it is advisable to cure the bonding resin separately, as it may counteract the polymerization shrinkage of the resin composite. During polymerization, an oxygen inhibited layer of 15 µm is formed over the adhesive resin. The unpolymerized resin present in this layer offer co-polymerization with the subsequently placed restorative resin. A thin uniform layer of adhesive resin is critical to bonding as it facilitates absorbing stresses of polymerization shrinkage. Use of air stream on adhesive resin should be avoided, as it may lead to uneven thinning of the bonding layer.

Mechanism of Dentin Bonding

The basic chemical composition of the dentin bonding adhesives is illustrated in Fig. 18.11.

The adhesive molecule is bifunctional, one part of which (X) enters into chemical union with the tooth structure, and the other part (M) copolymerizes to the resin through the double bond of methacrylate. Adhesion with dentin can occur either to the inorganic or organic components present on the surface of the

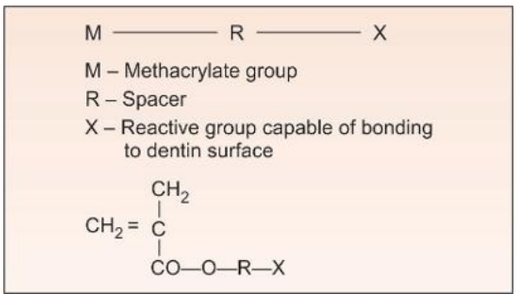


Fig. 18.11: Chemistry of dentin bonding agent

tooth. The spacer group (R) is responsible for making the molecule large enough to keep the methacrylate groups spatially located for optimal chemical reaction with the composite. Ideally, dentin adhesives should be both hydrophilic and hydrophobic. It is required to be hydrophilic so as to displace dentinal fluid and wet the surface, permitting penetration into porosities and eventually react with organic/inorganic components. Hydrophobic properties allow bonding to the composite resin (the matrix of composite is hydrophobic in nature).

Bonding to Inorganic Portion of Dentin

Bonding to the inorganic part of dentin involves ionic interaction between positive Ca^{++} ions on the surface of dentin and negative charges on the group X of the adhesive. Group X can be phosphates, amino acids and amino-alcohols or dicarboxylates. The bonding mechanisms for all the three are presented in Fig. 18.12A to C. To the left, Ca^{++} ions are seen and to the right are adhesive molecules. Agents that use a phosphate group in their bonding to calcium ions are referred to as phosphate bonding systems and are the most common ones employed. The substituent 'Z' in the phosphate-based adhesives may be chlorine, a hydroxy or a phenyl group.

The bond strengths of calcium bonding adhesives when used alone, seldom exceed 6 MPa. However, when different Ca^{++} bonding adhesives are combined or used in conjunction, bond strength may rise to 10–15 MPa. Examples of adhesives that involve bonding to inorganic portion of dentin are: Bondlite, Scotch bond, J&J Bonding agents, Clearfil, Prisma Universal Bond, etc.

Bonding to Organic Portion of Dentin

Bonding to the organic part of dentin involves interaction with the amino (–NH), amido (–CONH), hydroxyl (–OH) or carboxylate (–COOH) groups present in the collagen of dentin. Removal of hydrogen from any of these groups allows combination with chemicals present in dentin

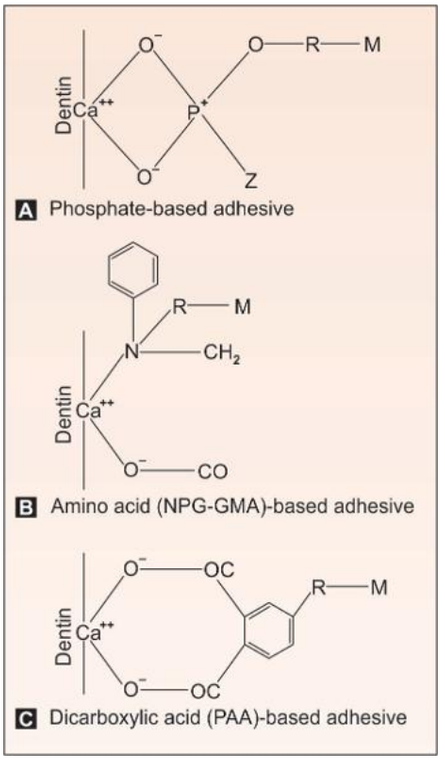


Fig. 18.12: Bonding to inorganic portion of dentin

bonding agents. Compounds that have a capacity to react with one or more groups of collagen are isocyanates, carboxylic acid chlorides, carboxylic acid anhydrides and aldehydes (Fig. 18.13).

Examples of the adhesives that rely on bonding to organic part of dentin are: Dentin Adhesit (isocyanate based), Gluma (aldehyde based), etc. Use of adhesives based on the concept of infiltration into moist collagen network to provide mechanical retention has started gaining importance than their ability to form chemical bonds to the tooth surface. The key to adhesion is to use a hydrophilic monomer that could easily infiltrate the collagen meshwork produced by conditioning of dentin (Fig. 18.14A, B).

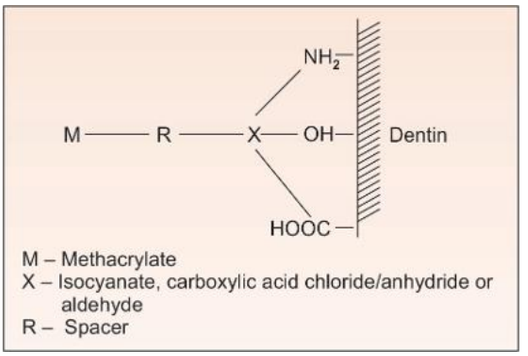


Fig. 18.13: Bonding to organic part of dentin

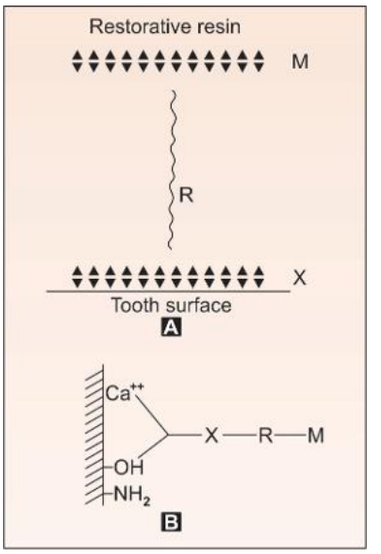


Fig. 18.14: Mechanism of bonding: **(A)** Bonding mechanism with intervening adhesive; **(B)** Bonding to inorganic/organic portion of dentin.

CLASSIFICATION (EVOLUTION) OF DENTIN BONDING AGENTS

For convenience, the evolution of dentin bonding agents have been classified as 'generations'.

1. First Generation Dentin Bonding Agents

Initially, a surface active co-monomer, N-phenyl-glycine-glycidyl methacrylate (NPG-GMA), was used, which acted as a primer between enamel/dentin and resin material by chelating with dentin (Fig. 18.15). Since calcium ions of the tooth substance are a mediator in the bond formation, agents of this type are expected to form stronger bonds to enamel than to dentin. Studies with this system have not shown good retentive values.

Cervident (1st generation) has shown 2.8 MPa bond strength to dentin and no improvement in marginal leakage when compared to conventional unfilled bonding agent.

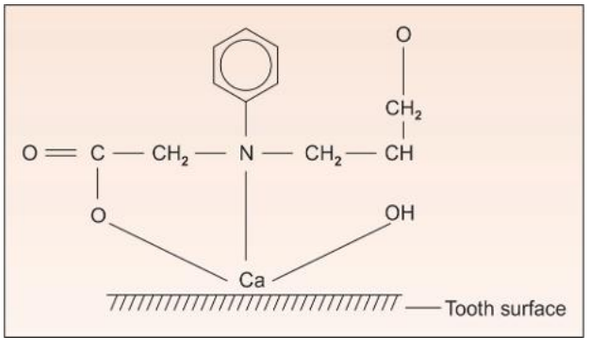


Fig. 18.15: Bonding of NPG portion to Ca^{++} ions by chelation

Drawbacks

- Showed poor clinical results
- Poor bond strength (2–3 MPa)

Examples

- Cervident
- Cosmic bond

2. Second Generation Dentin Bonding Agents

The second generation bonding agents were phosphorus/chloro phosphorus esters of unfilled resins, such as BisGMA or HEMA. The bonding mechanism involved ionic interaction between the phosphate group and calcium of the tooth. Clearfil (Kuraray, Japan) was the first agent introduced in this series. It composed of an ethyl alcohol solution containing tertiary amine as the activator. The catalyst liquid was BisGMA monomer containing a phenyl phosphate ester, benzoyl peroxide and methyl methacrylate. The interfacial bond was established through attraction between the negative charges of the oxygen on the phosphorus group and the positively charged calcium ions in the dentin surface (Fig. 18.16).

Scotchbond (phosphorus ester of BisGMA) is formed by reaction between BisGMA and phosphorus oxychloride ($POCl_3$). Bonding to tooth calcium is executed through chlorine having a partial negative charge (Fig. 18.17). Another explanation of its mode of action is that the chloro phosphate group becomes rapidly hydrolyzed on contact with moisture at the dentin surface producing a reactive group and liberating hydrochloric acid. This acid probably plays a part in bond formation by altering the structure of surface dentin including the smear layer.

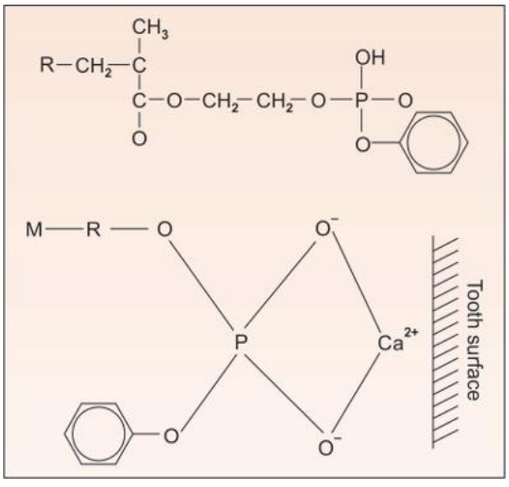


Fig. 18.16: Bonding mechanism in Clearfil (second generation)

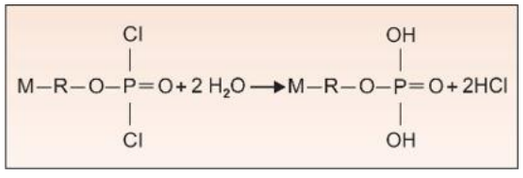


Fig. 18.17: Bonding mechanism in Scotchbond (second generation)

The bond strength have been reported to be 2–7 MPa for the second generation bonding agents; considered 'weak' to counteract the polymerization shrinkage of composite resins. Furthermore, hydrolysis of the bond between phosphonate esters and dentin, has been reported.

Drawbacks

- Lack of adequate bond strength that could overcome contraction stresses during polymerization.
- Being hydrophobic in nature, close adaptation to the hydrophilic dentin could not be achieved.
- Biocompatibility was not appropriate.
- Lack of sufficient knowledge about the presence and nature of smear layer. The adhesive bonded to the smear layer, rather than the dentin. As a result, the bond achieved was limited by cohesive failure in the smear layer or a break at the smear layer–dentin interface.

  Examples
  - Sctochbond
  - Clearfil Bond
  - Dentin Adhesit

3. Third Generation Dentin Bonding Agents

As further research ensued, it became evident that the smear layer had a negative influence on the performance of adhesive systems. To overcome this, an additional step (conditioning and priming dentin) was employed to either modify or remove the smear layer before applying the adhesive. The bond strength to dentin with these agents were comparatively better (7–15 MPa) and more durable; subsequently reducing microleakage.

The three steps—conditioning, priming and bonding may be carried out either separately or the components combined to reduce it to two steps. The conditioner is an acidic solution that removes the smear layer and is rinsed off after application. The acid opens up the dentinal tubules partially and increases their permeability. The residual acid must be rinsed off completely before primer is applied. The primer solution usually contains an adhesion promoter in a solvent, such as water, ethanol or acetone. These are applied to the surface and dried, presumably leaving the adhesion promoter on the

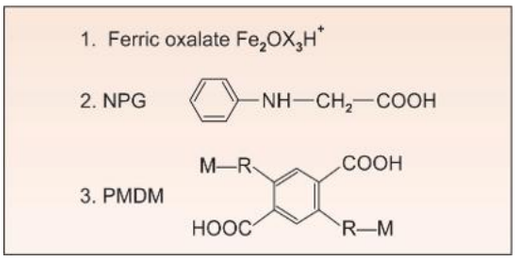


Fig. 18.18: Three-step oxalate bonding system (third generation)

dentin with its hydrophobic groups exposed to create a favorable surface for the bonding agent.

First system of the third generation, known as the oxalate bonding system (Fig. 18.18), utilized a solution of acidic ferric oxalate (2.5% nitric acid + ferric oxalate) on the enamel and dentin surface (act as conditioner). The ferric ions are adsorbed onto the dentin to increase the number of positive ions on the surface. This is followed by the application of an acetone solution of NTG-GMA (reaction product of N p-toluidine glycine and glycidyl dimethacrylate) and then another acetone solution of PMDA (pyromellitic dianhydride + 2 HEMA). This technique improved the resin bond. 'Tenure' was the first commercial oxalate bonding system, which utilized phosphoric acid in conjunction with aluminum oxalate and nitric acid as conditioner. Later, ferric oxalate was replaced with aluminum oxalate, to eliminate the possible risk of discoloration due to ferric ions. Another agent, Gluma (Fig. 18.19) utilizes 0.5M EDTA to remove the smear layer.

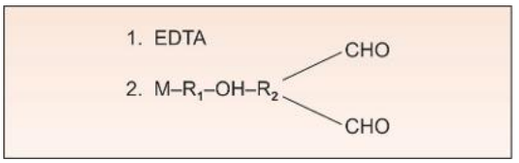


Fig. 18.19: Gluma bonding system (third generation)

Drawbacks

- Clinical studies showed decrease in retention with time (the longevity of the bond not maintained).
- Technique sensitive
- Time consuming
- Less bond strength

Examples

- Tenure
- Gluma
- Scotchbond 2
- Universal bond 2

4. Fourth Generation Dentin Bonding Agents

To improve upon bonding to dentin, Nakabayashi (1982) proposed a concept in which the resin is impregnated into partially decalcified dentin followed by polymerization, creating a resin reinforced layer, commonly known as *'hybrid layer'*. Hybrid layer is defined as *"the structure formed in the dental hard tissues by demineralization of the surface and subsurface followed by infiltration of monomers and subsequent polymerization"*. The mineralized tissues of peritubular and intertubular dentin are dissolved by acidic action; the initial surface penetration exposes the collagen fibers.

It has been suggested that excessive drying causes collapse of the collagen meshwork and prevents penetration of the subsequently applied primer and resin. A thin film of water presents on the conditioned dentin helps suspend the collagen fibrils and creates space for penetration. The earlier product, All Bond-2, uses an etchant of 35% phosphoric acid, followed by application of hydrophilic primer containing 2.0% NTG-GMA (N-tolyglycine-glycidyl methacrylate) and 16% BPDM (biphenyl dimethacrylate) in ethanol or acetone.

Subsequently, an unfilled resin containing BisGMA and HEMA is applied. Mean bond strength of 21.4+7.8 MPa can be achieved with fourth generation bonding agents.

Scotchbond Multi-Purpose, another agent, uses 10% maleic acid as etchant and aqueous solution of HEMA and polyalkenoate copolymers as primer. A few clinicians prefer phosphoric acid to etch enamel and maleic acid to condition dentin. Use of 35% phosphoric acid to etch both enamel and dentin has also been advocated. The bond strength achieved was 21.0 MPa with wet dentin and 18.0 MPa with dry dentin.

Advantages

- Reduced technique sensitivity.
- Similar bond strength to enamel and dentin.
- No reduction in bond strength when applied to moist surface or under conditions of high humidity.
- Some systems can bond to mineralized tissue as well as metal, amalgam, porcelain and indirect composite restorations.

Examples

- All Bond-2
- OptiBond B and C
- Scotchbond Multi-Purpose
- Metabond

5. Fifth Generation Dentin Bonding Agents

The fifth generation adhesives are also based on the concept of hybridization, relying on wet bonding technique. It differs from its predecessor in that it uses one component resin, i.e. after conditioning of enamel and dentin, the steps of priming and bonding are combined so that bonding is achieved with a one component formula.

Prime and Bond NT and Prime and Bond 2.1 contain PENTA, TEGDMA and an elastomeric UDMA resin in acetone (Fig. 18.20A, B). PENTA is a multifunctional molecule, helps to partially demineralize dentin, facilitating penetration of resins.

The system is highly sensitive to even mild desiccation of acid conditioned dentin. These adhesives do not necessitate dentin conditioning, but require acid conditioning on enamel for optimal enamel bonding.

OptiBond Solo (Kerr) is another one component adhesive containing HEMA, GPDM and BisGMA in an ethanol/water system (improved version of optiBond). It also contains fillers, like fumed silica and barium, but at half the concentration of that found in OptiBond.

Single Bond (3M) is also one component adhesive containing HEMA, BisGMA, dimethacrylate resin and a unique methacrylate functional copolymer of polyacrylic and itaconic acids in water and ethanol solvent base. Bond strength for fifth generation systems are almost equal to those of fourth generation agents, i.e. 17–24 MPa. Although these systems have reduced the number of components, they are not necessarily faster or simpler procedures. None of these agents is truly a one component system, because almost always they require a pretreatment with acid on enamel and sometimes on dentin. In addition, these materials are sensitive to even minor drying and may require multiple applications of the adhesive to ensure optimal penetration of resin.



Fig. 18.20: Fifth generation bonding agents: **(A)** Prime and Bond NT (the priming and bonding steps are combined); **(B)** Prime and Bond 2.1

Advantages

- Bond strength is sufficient.
- Postoperative sensitivity is rare.
- Some agents have incorporated fluoride and elastomeric components to improve marginal integrity.
- Time saving and simple to use.
- Elimination of washing out of the acidic gel, therefore, elimination of risk of collagen collapse.
- Combination of etching and priming steps decreases the working time.

Disadvantages

- Solution must be refreshed continuously.
- Residual smear layer may remain between the adhesive material and dentin.
- Not essentially a single component system.
- Multiple coats are required.

Examples

- OptiBond Solo
- Single Bond
- Prime and Bond NT
- Prime and Bond 2.1

6. Sixth Generation Dentin Bonding Agents

To improve upon the bond strength and to facilitate manipulation, this generation of adhesives relied on the use of acidic primer (pH 2.5–4.5), eliminating the need for phosphoric acid as etchant (Fig. 18.21). The manipulation is easy and a sufficient bond can be achieved in dentin; however, bond with enamel is less effective. This may be due to the fact that the sixth generation systems are composed of an acidic solution that cannot be kept in place for long and must be refreshed continuously. Moreover, pH of these systems is not adequate to properly etch enamel.



Fig. 18.21: Sixth generation bonding agent (self-etch adhesive)

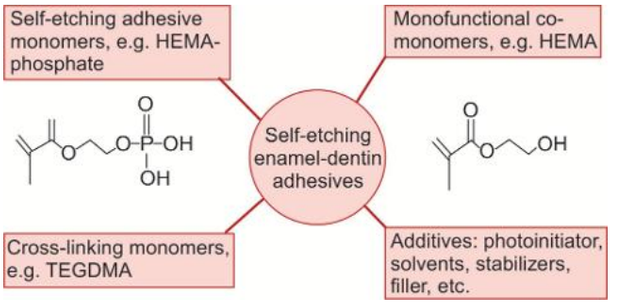


Fig. 18.22: Components of self-etching adhesives

The diagrammatic representation of components of self-etching adhesive is depicted in Fig. 18.22.

Sixth generation dentin bonding agents are further of two types: Type 1 (self-etching primer and adhesive) and Type 2 (self-etching adhesive).

Type 1 (Self-Etching Primer and Adhesive)

These are available in light cure or dual cure product. Self-etching primer and resin adhesives are applied in separate layers.

- Two bottles [Liquid 1 (acidic primer), Liquid 2 (adhesive)]; acidic primer applied to tooth followed by adhesive.
- Unprepared enamel may require etching with phosphoric acid.
- Light-cure/dual-cure formulations
- Solvent is water.

  Examples:
  - NanoBond
  - OptiBond
  - Prompt L-PoP (Fig. 18.23)



Fig. 18.23: Prompt L-PoP (sixth generation adhesive)

Type 2 (Self-Etching Adhesive)

These are available only in light cure type. The self-etch primer and adhesive is mixed outside and applied.

- Two bottles (containing acid primer and adhesive); a drop of each liquid is mixed and applied to tooth.
- Unprepared enamel may require etching with phosphoric acid.
- Light-cure formulations
- Solvent is water

  Example:
  - Xeno III (Fig. 18.24)

Advantages

- Designed to be used on dry dentin.
- Do not etch very far into dentin beneath smear layer (avoids removal of smear plugs).

7. Seventh Generation Dentin Bonding Agents

Seventh generation adhesives are available both as light cure and dual cure types. All three components are combined in a single bottle.

These bonding agents are slight modification of type 2 sixth generation bonding agent (Fig. 18.25).



Fig. 18.24: Xeno III (sixth generation self-etch adhesive)



Fig. 18.25: Light cure (seventh generation) adhesives

The features are:

- More acidic (pH <1)
- Single bottle containing acidic adhesive.
- Unprepared enamel may require etching with phosphoric acid.
- Solvent is water.

Disadvantages

- Relatively short shelf life (acidity of maleic acid accelerated the breakdown of HEMA).
- Hydrophilic (attract and absorb water leading to leaching of unpolymerized monomers or hydrolytic degradation products through water filled channels called 'water trees').

Examples:

- G-Bond
- i-Bond

8. Eighth Generation Bonding Agent

A modified version of seventh generation bonding agent is introduced by VOCO America as Futurabond DC. This is one-step dual-cure, non-filled, self-etch adhesive, available in single use blister packs. The 'chemical cure' mode makes the product suitable to be used in root canals, avoiding blocking the root canals with cured bonding agent.

Advantages

- Can be used with light-cure/dual-cure/self-cure composites.
- Provides high bonding strength.
- Moisture tolerant.
- Contains fluorides.
- Nanofillers help in better crosslinking of the bonding resin components.

Example:

- Futurabond

Table 18.2: Generations of dentin bonding systems

1st generation	Primers with low bond strength
2nd generation	Dentin-enamel bonding agents with improved bond strength to etched enamel
3rd generation	Etching of dentin and partially removal of smear layer
4th generation	Total-etch technique and formation of the hybrid layer and resin tags
5th generation	Simplification of clinical procedures: One-bottle systems and self-etching primers
6th generation	One-step bonding systems with bonding to enamel and dentin; Type 1 are compatible with self-cured composites and Type 2 are not
7th generation	Single bottle, no mix, self-etching adhesive; not compatible with self-cure composites

The generations (evolution) of dentin bonding system are summarized in Table 18.2.

Alternative Classification (Evolution) of Bonding Agents

Since the generations are becoming confusing, an alternative classification (currently used) is presented, which involves strategy of etching and rinsing along with steps followed in bonding procedures.

I. *Etch and rinse adhesive*

A. Three steps

- Acid
- Primer
- Bonding agent

Examples

- All-Bond 2 (Bisco)
- Clearfil Liner Bond (Kuraray)
- Scotchbond Multi-Purpose (3M ESPE)
- OptiBond DC (Kerr)

B. Two steps

- Acid
- Primers + Bonding

Examples

- Gluma (Heracus-Kulzer)
- One-Step (Bisco)
- Prime and Bond (Dentsply)
- Syntac (Ivoclar)

II. *Self-etch adhesive*

A. Two steps

- Acid + Primer
- Bonding agent

Examples

- Clearfil Liner 2 (Kuraray)
- Clearfil SE (Kuraray)

- Prisma Universal (Dentsply)
- Solobond Plus (VOCO)

B. One step
- Acid + Primer + Bonding agent

Examples
- Clearfil 3 (Kuraray)
- i-Bond (Heraeus-Kulzer)
- Futurabond (VOCO)
- Xeno III (Dentsply)

III. *Universal (multimode) adhesive*

I. *Etch and Rinse Adhesives*

The initial three-step adhesion system involves acid etching, priming and application of adhesive (three steps). A hybrid layer is formed with relatively hydrophobic adhesive layer.

In etch and rinse adhesives, initial step 'etching' is followed by rinsing, which removes smear layer and smear plugs. Acid etching leads to demineralization of dentin up to a depth of 3.0–5.0 μm, exposing a scaffold of collagen fibrils (no hydroxyapatite). In priming, hydrophilic monomer, such as HEMA, dissolved in organic solvents (ethanol, acetone or water), is applied. HEMA improves wettability and promotes expansion of collagen network; solvents displace water from the dentin surface, facilitating subsequent resin infiltration.

In bonding step, solvent-free adhesive resin is applied, which penetrates dentinal tubules and interfibrillar spaces of collagen network. Upon polymerization, a hybrid layer is formed at dentin–restoration interface. From these three-step etch and rinse adhesives, two-step adhesives were developed; combining primer and adhesive. The two-step adhesives present reduced ability to infiltrate the demineralized dentin. The solvents also do not evaporate properly and may get entrapped within the adhesive layer after polymerization. The over-decaying of dentin may hamper the monomer diffusion and subsequently the functional hybrid layer. These effects decrease the mechanical quality of hybrid layer; may lead to its early degradation. The substrate should not be over-dry or over-wet for the longevity of bonding. The main drawback of etch and rinse system is the nanoleakage, which affects durability of the bonding.

II. *Self-Etch Adhesives*

To minimize nanoleakage and to control sensitivity to humidity in etch and rinse adhesives, self-etch adhesives were developed. The pH of acidic component in self-etch adhesive is relatively higher than phosphoric acid; three variants are available—mild, moderate and strong, depending upon pH of the acidic component. At enamel, 'strong' self-etching shows good results; whereas, mild is ineffective.

Self-etch adhesives are categorized as one-step or two-step adhesive, depending upon the number of clinical applications. One-step self-etch adhesive combines the etching, priming and bonding (acidic, hydrophilic/hydrophobic monomer and organic solvents in one single solution). Two-step self-etch adhesives include use of hydrophobic etching primer (acidic monomer, which etch and prime the substrate), followed by application of bonding agent.

Self-etch adhesives are highly hydrophilic and attract water, which may increase the potential for degradation. Water is an essential ingredient because it ionizes the acidic group, facilitating etching of hydroxyapatite (self-etch adhesive contains 30–40% water).

One-step self-etch adhesives when applied to hydrated dentin, absorb and retain water, behaving the semipermeable membrane. These water fluxes are responsible for water-filled channels, within the adhesive layer, known as 'water trees' (described separately in Chapter).

Two-step self-etch adhesives contain two separate components—first primer plus acid, etc. and second hydrophobic bonding resin. Self-etch adhesives reduce the possibilities of mix-manipulation during acid etching, rinsing and drying, which occurs with etch and rinse systems.

Certain drawbacks have been observed with the use of self-etch adhesives. One drawback is their inability to etch enamel to the same depth as phosphoric acid does; secondly, the acidic content may increase the hydrophilicity of the adhesive layer, leading to water uptake and subsequently degradation of bond. (One-step adhesive is inferior in terms of bond durability.)

The use of individual component in self-etch adhesive system is summarized in Table 18.3.

Table 18.3: Use of individual component in self-etch adhesive systems

UDMA/HEMA	• Conditioning of enamel and dentin wetting agent, helps in thin film formation • Promotes infiltration
4-META	• Binding to calcium of apatite • Binding to collagen
Glutaraldehyde	• Disinfectant • Desensitizing agent
Acetone	• Helps in removing humidity • Solvent for monomer
Water	• Helps in etching process • Solvent for monomer

III. *Universal (Multimode) Adhesives*

The universal, multimode or multipurpose adhesives are basically improved version of self-etch one-step adhesives (Fig. 18.26), which can be applied on etched or unetched enamel and dentin (used as self-etch adhesive on dentin and etch and rinse adhesives on enamel). These bonding agents are indicated to be used as silane for glass ceramics and primers for metal alloys.

Advantages:

- Universal application
- Effective both on wet and dry dentin surfaces.

The current classification of adhesive system is summarized in Table 18.4.

Composition of commonly used bonding systems and their advantages and disadvantages are presented in Tables 18.5 and 18.6, respectively.

CATEGORIZATION OF ADHESIVES BASED ON THEIR ACTION ON SMEAR LAYER

The dentin adhesive systems are also categorized on their clinical approach to smear layer. Based on this criteria, the adhesives are of three types:

a. ***Type 1 (modifying smear layer):*** The adhesives modify the smear layer and incorporate it into the bonding process. Their application can be completed in one or two steps, using a single adhesive or primer and adhesive, respectively. The smear layer in this adhesion process is retained based on the concept that it serves as a natural barrier to the pulp, prevents bacterial invasion and limits the outward flow of pulpal fluid that might impair bonding efficiency. Penetration of monomers into the smear layer and their subsequent polymerization reinforces the attachment of the smear layer to the underlying dentin and also forms a micromechanical and perhaps a chemical bond to the underlying dentinal surface. It is, however, observed that resin is able to penetrate only to a limited extent into the superficial dentin. Clinically, these systems require selective etching of enamel in a separate step.

b. ***Type 2 (removing the smear layer):*** The adhesives completely remove the smear layer and are subdivided into two- and three-step application. A two-step process involves dentin conditioning followed by a combined primer and adhesive; whereas, the three-step process involves separate conditioning, priming and bonding agent application. Any of the processes results in complete removal of the smear layer by acidic conditioners that are simultaneously applied to enamel and dentin using the total etch technique. The action of these agents is principally based on the combined effect of hybridization and resin tag formation.

c. ***Type 3 (dissolving the smear layer):*** The adhesives dissolve the smear layer rather than remove it. The process is accomplished in two steps using a combined conditioner and primer (self-etching primer) followed by application of the adhesive resin. The slightly acidic primers supplied with these systems partially demineralize the smear layer and the underlying dentin surface without removing the dissolved smear layer remnants and unplugging the tubule orifices. The rationale behind their use is to superficially demineralize dentin and simultaneously penetrate it with



Fig. 18.26: Universal adhesives

Table 18.4: Current classification of adhesive systems

Classification	*Acid*	*Primer*	*Bonding resin*
Etch-and-rinse, 3 steps	Phosphoric acid 32–40%	Hydrophilic monomers, organic solvents	Hydrophobic monomers
Etch-and-rinse, 2 steps	Phosphoric acid 32–40%	Hydrophilic monomers, organic solvents	Hydrophobic monomers
Self-etch, 2 steps	Acidic and hydrophilic monomers	Organic solvents	Hydrophobic monomers
Self-etch, 1 step	Acidic and hydrophilic monomers	Organic solvents	Hydrophobic monomers

Table 18.5: Composition of commonly used dentin adhesive systems

Adhesive system	*Etchant/self-etching primer*	*Adhesive*
Single Bond	35% phosphoric acid gel	BisGMA, HEMA, polyalkenoic acid copolymer, ethanol, water, photoinitiator
Clearfil Liner Bond 2	Primer A (Phenyl P, 5-NMSA, CQ, Ethanol) Primer B (HEMA, water)	MDP, HEMA BisGMA, microfiller
One Step	32% phosphoric acid gel	BPDM, BisGMA, HEMA, acetone, photoinitiator
Imperva Fluoro Bond	Primer A (Water, acetone, initiator) Primer B (4-AET, HEMA, 4-AETA, initiator)	4 -AET, HEMA UDMA, glass-ionomer Filler, microfiller
Prime & Bond NT	34% phosphoric acid gel	Di & Trimethacrylate resins, PENTA, acetone
OptiBond Solo	37.5% phosphoric acid	BisGMA, HEMA, ethyl alcohol
Scotchbond Multi-Purpose	HEMA, Polyacrylic acid, water	BisGMA, HEMA
Liner Bond 2V	Primer A (MDP, HEMA, water, photoinitiator, accelerator) Primer B (HEMA, water accelerator)	MDP, dimethacrylates, photoinitiator, accelerator, microfiller
Clearfil SE Bond	MDP, HEMA, Hydrophilic dimethacrylate di camphorquinone N-N-Diethanol-p-toluidine water	MDP, BisGMA, HEMA, hydrophobic di-camphorquinone dimethacrylate N-N-Diethanol-p-toluidine silanated colloidal silica
Prompt L-Pop	Self etching	Water methacrylated phosphoric oxide fluoride complex with zinc photoinitiators
Tyrian, SPE	Self etching primer [2-acrylamido 2-methyl propane sulfonic acid BIS (methacryloyloxy ethyl)] phosphate ethanol	Biphenyl dimethacrylate EMA, acetone glass filling photoinitiators
Reactmer Bond (RB)	Bond A Bond B	4-MET, 4-META HEMA, UDMA, water, photoinitiator, fluoride complex
XENO III	Universal catalyst	METP, UDMA, HEMA, ethanol, water, silica, filler, photoinitiator, fluoride complex
OB2	Bond A Bond B	MDP-10, HEMA Phosphoric acid ester, photoinitiator, water, aluminosilicate glass filler
One up Bond F	Self-etching	HEMA, MMA, methacrylolyoxy alkyl acid phosphate, fluorosilicate glass filler, water, photoinitiator
i-Bond	Self-etching	Acetone/Water UDMA, 4-META, Glutaraldehyde

PENTA – dipentaerythritol pentaacrylate monophosphate
MDP-10 – Methacrylolyoxy decyl dihydrogen phosphate
BisGMA – Bisphenyl glycidyl methacrylate
NMSA – N-methacryloxy-5-amino salicylic acid
HEMA – Hydroxyethyl methacrylate
Phenyl P – 2- methacryloyloxy ethyl –phenyl hydrogen phosphate
BPDM – Bisphenyl dimethacrylate
4 AET – 4-acryloxyethyl trimellitic acid
CQ – Camphorquinone
4 AETA – 4-acryloxyethyl trimellitate anhydride

Table 18.6: Advantages and disadvantages of commonly used bonding agents

Bonding agents	*Advantages*	*Disadvantages*
i. Self-etching primer (Clearfil SE Bond) Ep + B	• No rinsing, quick application • Less postoperative sensitivity • Clinical studies support their use on dentin • Bond very well to dentin etched with phosphoric acid	• May result in enamel microleakage due to deficient enamel etching • Slight degradation of the hybrid layer
ii. Self-etching all-in-one adhesive (Prompt L-Pop) Ep + B	• No rinsing, very quick applications • Results in enamel etch pattern similar to that of phosphoric acid • No bottles, no cross-contamination	• Has resulted in a wide range of bond strength values • Bonds better with compomers than with composites • May need multiple coats to bond effectively to dentin • Does not bond well to unprepared enamel • Not indicated for indirect restorations
iii. Total etch multi-bottle adhesives (All-Bond 2, Scotchbond Multi-Purpose, OptiBond) E + p + B	• Clinical research favour their use on different substrates • Can achieve highest dentin bond strengths among all the adhesives • Generally contain a dual-cure option for indirect restorations and bonded amalgams	• Use of multiple bottles is cumbersome • Possibility of running out of Primer A before Primer B (or vice-versa) • Because primer and adhesive resin are dispensed into the same plastic container, their sequential application may be inverted
iv. Total etch one-bottle adhesives (Excite, One-Step, OptiBond SOLO Prime & Bond NT) E + pB	• Laboratory research favour their use on enamel and dentin • Clinical studies observed positive results • One-bottle concept is user-friendly	• Generally, lower bond strengths than multi-bottle adhesives, acetone-based adhesives may lose their efficacy with constant utilization • Acetone-based adhesives may need more coats than those recommended by the manufacturer (*rule of thumb: to prevent the occurrence of dry spots not covered with adhesive, always apply one extra coat when using acetone-based adhesives*) • Ethanol-based adhesives may pool easily around the preparation margin • Most one-bottle adhesives cannot be used as dual-cure materials • Have potential for excessive dentin decalcification; rely on careful moisture control to achieve good bonding

(E–Etching, p–priming, B–bonding agent)

monomers, which can be polymerized *in situ*. The adhesives based on their action on smear layer are summarized in Table 18.7.

RESIN–DENTIN BOND: DEGRADATION AND MANAGEMENT

The dentinal adhesives are generally vulnerable to degradation. The phenomenon of degradation is different for 'etch-and-rinse adhesive' and 'self-etch adhesive'. Researchers have been trying to devise techniques/materials to stabilize dentin–resin bonds, since long. Let us understand why the bond gets degraded with time.

i. ***Etch-and-rinse adhesives:*** After rinsing the etching solution, the spaces between the collagen fibrils get over filled with water. The monomer easily penetrates into this hydrated dentin surface forming a receptive hybrid layer. (Drying of dentin prevents infiltration of monomer into the nanochannels of collagen fibrils.) The wet-bonding technique with etch-and-rinse adhesive have shown promising results. (Water preserves the porosity of collagen network for monomer diffusion.)

The interfibrillar spaces of collagen matrix may utilize most of water for collagenolytic enzymes activity. The residual water in the collagen can result in separation of resin adhesive–dentin interface, by weakening the polymer within hybrid layer; subsequently, making it susceptible

Table 18.7: Categorization of adhesives based on their action on smear layer

Systems	*Smear layer modifying*	*Smear layer removing*	*Smear layer dissolving*
One step	• Hytac OSB • Pertac Universal Bond • Prime & Bond 2.1 • Solist • Tokuso Light Bond		
Two-step	• Optec Universal bonding • Pro bond • Tokuso Light Bond (two-step) • Tripton	• Fuji Bond LC • Gluma 2000 • OptiBond Solo • Prime & Bond 2.0 (total etch) • Scotchbond 1 (Single bond) • Syntac Single component	• Clearfil Liner Bond 2 • Denthesive 11 • OptiBond (no etch) • Imperva Bond (no etch) • Scotchbond 2 • Syntac • XR bond
Three-step		• All Bond 2 • Amalgam Bond Plus • Clearfil Liner Bond • Gluma • Imperva Bond (total etch) • Mirage Bond • OptiBond (total etch) • Scotchbond Multipurpose • Scotchbond Multipurpose Plus • Tenure	

to degradation by enzymes. Along with hydrolytic degradation of adhesive resins, proteolytic degradation of collagen fibrils are also responsible for decline in dentin-bond strength.

ii. ***Self-etch adhesive:*** Self-etch adhesives are hydrophilic (attract water), which increases the potential for degradation. Water is an important ingredient of self-etch primers; water ionizes acidic groups and etch hydroxyapatite. Water also facilitates solubilization of the byproducts resulting after etching process. Although water is essential for ionization of acidic resin monomers, it may prevent formation of a strong polymer within the hybrid layer. (One step self-etch adhesive absorbs higher quantity of water than two-step self-etch adhesive.) The hydrophobic to resin layer can make the two-step self-etch adhesives more impermeable to water sorption; improving the longevity of clinical bond strength. In water-tree phenomenon, water that migrates to adhesive–composite interface will be trapped by the overdrying hydrophobic composites, forming water blisters. These water blisters result in mechanical disruption of bond between adhesive and composite resin. The higher hydrophilicity of self-etch adhesives, especially the one step self-etch solutions, make the bonding vulnerable to water sorption. The hybrid layer may also get exposed by endogenous proteases. (MMP inhibitors to self-etch primers prevent resin-dentin bond degradation.)

Stabilizing Resin–Dentin Bond

The methacrylate-based monomers (main constituent of dental adhesives), and their hydrolytically susceptible groups, viz. esters, hydroxyl, urethane, etc. usually compromise with the integrity of the adhesive layer. The degradation mechanism of bond (collagen and resin monomer) and prevention is summarized in Table 18.8. The continued research to increase and stabilize the bond between resin-based materials and the dental substrates has led to development of newer agents; a few bonding protocols have also been suggested.

1. *Collagen Crosslinking Agents*

A few external crosslinking agents can effectively reinforce collagen fibrils through intermolecular cross-linking; stabilizes bond by reducing molecular mobility. Certain oxidizing agents (sodium hypochlorite, hydrogen peroxide, etc. used in various clinical situations) may hamper polymerization of resin-based materials. Antioxidants are being tried to eliminate any oxidizing species from the dental structure (antioxidants improve bond strength between resin materials and the substrates that underwent an oxidizing process). Use of protease inhibitors [matrix metalloproteinase (MMP) and

Table 18.8: Degradation mechanism of bond (collagen and resin monomer) and prevention

Degradation mechanism	*Prevention*
Collagen	
• Hydrolysis by residual water in the adhesive and also the water not removed after rinsing the etchant • Exposed collagen from the unstable polymers in the hybrid layer (exposed collagen fibers show increased susceptibility to mechanical fatigue) • Collagenolytic enzymes digest unprotected collagen fibers	• Inhibitors of collagenolytic enzymes (epigallocatechin gallate) • Inhibitors of metalloproteins matrix • Use of hydrophobic bonding resins • Cross-linking agents (glutaraldehyde, carbodiimides, etc.) • Remineralization (biomimetic) • Quaternary ammonium inhibit metalloproteins
Resin monomer	
Degradation mechanism	*Prevention*
• Hydrolysis by residual water in the adhesive and also by dentin humidity • Impregnation of monomer into exposed collagen fibers • Self-etching primers are semipermeable membranes (water blisters may result in disruption of the bond) • Continuing etching of dentin as in self-etch adhesives	• Use of hydrophobic bonding resins and monomer with increased conversion rate • Initiators that facilitate polymerization of monomer, even in the presence of water • Use of resin monomer incorporated with antibacterial agents • Adhesive materials capable of setting incorporated water

collagen crosslinkers (CC)] has been recommended to prolong dentin-bond durability.

The common agents and their mode of action are summarized in Table 18.9.

2. *Reinforcement of Resin Matrix*

It is established that hybrid layer get degraded by enzymatic activity and chemical activity. Certain components, which are used to reinforce the dental adhesives, have shown advantageous effects.

Various fillers and nanoparticles have been tried adding to adhesives to achieve resistance to degradation. Zirconia particles when added to adhesive (Scotchbond multipurpose, etc.) have shown improved bond strength. As the hybrid layer becomes stronger, water uptake gets reduced, which diminishes the process of hydrolysis and protease activity; subsequently, reducing the chances of bond degradation. Silver and copper nanoparticles along with zinc oxide particles have been incorporated into adhesive components. Reinforcement of dental adhesives with fillers/nanoparticles has strengthened the hybrid layer and substantially improved the bonding characteristics (Table 18.10).

3. *Substrate Conditioning with Laser Prior to Bonding*

It is established that laser irradiation of the substrate (enamel or dentin) prior to bonding results in stable and enhanced bond strength. Erbium-doped yttrium aluminum garnet (Er:YAG) laser has been commonly used to condition dental substrates. Laser creates active groups, such as carboxyl/carbonyl groups onto dentin surface, improving wetting and chemical interaction of the adhesive monomers. Lasers have also shown promising results when applied to dentin submitted to oxidizing agents (Table 18.11).

4. *Modifying Bonding Protocol*

Earlier authors suggested using double coat of adhesive resin with the aim of producing longer resin tags and less chances of hydrolysis, thereby resulting in improved resin–dentin bonds. A hydrophobic resin has also been applied after the bonding step; the extra addition of hydrophobic resin leads to densely packed hybrid layer, which is less prone to long-term degradation.

In an another approach, the dentin is dehydrated with 50–100% ethanol for 10 seconds followed by blot drying the excess ethanol. The amount of residual water is reduced in the hybrid layer, improving the resin–dentin bond. The dentin has also been rinsed with distilled warm water followed by blot drying prior to bonding. The process increases free energy of the surface; subsequently, increasing wettability of the resin adhesives. The residual water gets eliminated from the hybrid layer, improving the resin-dentin bond (Table 18.12).

BONDING IN OTHER CLINICAL SITUATIONS

The spectrum of bonding is quite wide and involves almost every material in dentistry. (Bonding spectrum in dentistry is illustrated in Fig. 18.27.)

The successful bonding has been achieved between restorative materials and the tooth tissues and also between different restorative materials.

Table 18.9: Improving resin–dentin bonds: Common agents and their mode of action

Agent (mode of application)	*Mechanism(s) of action*
1. Glutaraldehyde (after acid etching)	Reducing collagen in the hybrid layer, and enhancing the cohesive strength of collagen fibrils (collagen crosslinking).
2. Carbodi-imide (after acid etching)	Form covalent bonds with collagen fibrils in the hybrid layer (collagen crosslinking).
3. Riboflavin (after acid etching)	Improve collagen matrix texture, facilitating adhesive resin penetration (collagen crosslinking).
4. Bromelin enzyme (after acid etching)	Improve surface energy, and enhance resin infiltration (collagen crosslinking).
5. Proanthocyanidin	
i. After acid etching	i. Improve collagen fibrils and enhance quality of the hybrid layer (collagen crosslinking).
ii. Sodium hypochlorite prior to bonding	ii. Restore the oxidized dentin and improve bonding (antioxidant effect).
iii. Acid etching along with priming	iii. Reduce collagenolytic activity and improve quality of the hybrid layer; subsequently, better bonding (protease inhibition).
6. Chlorhexidine	
i. After acid etching	i. Improve and strengthen the hybrid layer (collagen crosslinking).
ii. Before applying cement	ii. Inhibit catalytic activity, prevents binding of calcium ions to metalloproteins (protease inhibitor).
7. Polyphenols	
i. After acid etching	i. Produce stable bonds with collagen fibrils, improving hybrid layer (collagen crosslinking).
ii. During the priming step as a modified primer	ii. The anticollagenolytic effects prevent degradation of dentin collagen within the hybrid layer (protease inhibitor).
8. Vitamin C and vitamin E (during priming)	Improve and strengthen the hybrid layer (antioxidant effect).
9. Minocycline/doxycycline (after acid etching)	Inhibit catalytic activity, prevent binding of calcium ions to the metalloproteins (protease inhibitor).

Table 18.10: Improving resin–dentin bonds: Reinforcement of resin matrix

Agents added to adhesive resin	*Findings*
• Copper nanoparticles	• Provide antimicrobial activity
• Silver nanoparticles	• Improve wettability
• Titanium dioxide nanoparticles	• Improve hybrid layer
• Zinc oxide nanoparticles	• Preserve integrity of the hybrid layer
• Aluminosilicate	• Infiltrate into the resin tags, and reinforce the hybrid layer

Table 18.11: Improving resin–dentin bond: Substrate conditioning with laser prior to bonding

Substrate	*Findings*
• Enamel etched with laser	• Facilitates changes in the substrates (removal of the smear layer and evaporation of water); increases surface area required for effective bonding.
• Dentin conditioned with laser	• Improves wetting and chemical interaction of the monomers.
• Bleached dentin conditioned with laser	• Accelerates release of free radicals originated during bleaching, effectively enhance bonding.
• Dentin exposed to sodium hypochlorite and conditioned with laser	• Increases interactions of the collagen fibrils with hybrid layer, improving bonding.

Bonding of Glass-Ionomers to Hard Tissue

Glass-ionomers possess the property of adhering to enamel and dentin; and also to other reactive polar substrates, such as base metals. The primary mechanism in their bonding is 'chemical adhesion' rather than 'micromechanical' as seen in composites. (Acid etching/surface roughening procedures are of no use with glass-ionomers.) On placing the hydrophilic and highly ionic glass-ionomer cements on the tooth substrate, they compete successfully with water because of the multiplicity of carboxyl groups that form hydrogen bonds with the substrate. Water is either displaced or incorporated into the cement. Adhesion so obtained is permanent and resistant to water degradation. Also, the ionic polar bonds that attach glass-ionomer cement to the substrate can be re-established, if broken, in contrast to covalent chemical bonds that cannot be reformed.

Table 18.12: Improving resin–dentin bonds: Modifying bonding procedure

Modification in bonding steps	*Findings*
• Dentin rinsed with distilled water prior to bonding	• Increase free energy of the surface, and wettability of the adhesive.
• Dentin dehydrated by sequential use of 50% and 100% ethanol	• Facilitate evaporation of water from the interfibrillar spaces of collagen fibrils.
• Hydrophobic resin applied after bonding	• Hydrophobic resin decreases the relative concentration of unreacted monomer in the adhesive layer.
• Applying two layers of adhesive (double coating)	• Produce longer resin tags and also control hydrolysis.

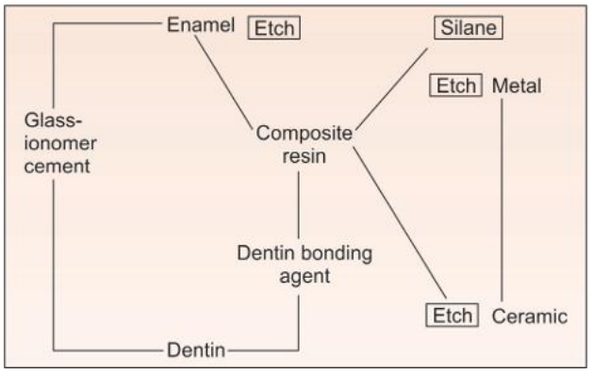


Fig. 18.27: Bonding spectrum in dentistry

In early stages of the chemical reaction, glass-ionomers are quite acidic, because of the presence of unreacted caryboxylic acid groups, which offer an adhesive mechanism with the Ca ions of hydroxyapatite ($Ca_{10}(PO_4)_6(OH)_2$) in enamel and dentin. As expected, bonding to enamel will be stronger than bonding to dentin because of the larger inorganic (hydroxyapatite) content in the former.

Initially, it was speculated that chelation of calcium in apatite by glass-ionomer was the primary mechanism involved in adhesion. The interaction between apatite and polyacrylic acid might have resulted in polyacrylate ions, which formed strong ionic bonds with the calcium ions in enamel and dentin surface. It is opined that when cement was initially applied to the tooth in a fluid consistency, wetting and adhesion occurred by hydrogen bonding mediated by carboxyl groups in the cement paste. As the cement set, hydrogen bonds were gradually replaced by ionic bonds. Later, it was postulated that a series of complex ion exchanges occur during chemical reaction and adhesion. As the polyacrylate ions enter the molecular surface of hydroxyapatite, they displace and replace one phosphate and one calcium ion. As a consequence, an intermediate layer of calcium and aluminum phosphate and polyacrylate is formed at the interface between the cement and apatite. Simply cations and anions may not be able to fill the gap at the interface; a polymer chain of cations and anions can bridge the gap and complete the bonding.

Bonding of Composite to Glass-Ionomer

Glass-ionomer is invariably used as a substitute for dentin under composite resins. This technique is commonly known as bilayered technique (Sandwich technique) in which the enamel and cement are etched prior to placement of the restorative resin.

Generally, 37% phosphoric acid is used to etch glass-ionomer and the enamel. Acid treatment of glass-ionomer improves its bond to composite by producing a rough surface (glass particles stand out above the matrix). Resin monomer penetrates into the micropores between the particles, thereby providing mechanical interlocking.

Low viscosity resins are advisable as they have a lower contact angle and better wetting ability with the substrate. The cement should be allowed to fully set before it is etched otherwise the unreacted particles may dissolve and weaken the cement. (Delay of 20 minutes is recommended.) Acid-etching should be restricted to 15–20 seconds that will remove enough matrix material to provide mechanically retentive surface without harming the strength of the underlying cement. Above 30 seconds of etching, the cement is excessively prone to degradation by the acid. Grinding of the set cement should be avoided, as it might decrease the bond strength values.

The type of glass-ionomer cement also influences the bond at the dentin surface and the glass-ionomer–composite interface. Higher the strength of the cement, better are the clinical results. The rate of set of the cement also affects bond strength. A slow setting cement will be weakened considerably, if etched prematurely. Faster setting cement can be etched after 2–5 minutes. Glass-ionomer cements should be used in adequate bulk under composite resin to avoid stresses from the shrinking composite. Use of light cure glass-ionomer lining materials have greatly reduced the chances of debonding. These achieve high early strength on photo-polymerization and also chemically bond to the resin composite without the need for etching. Should a layer of separating medium be applied at the glass-ionomer composite interface, remains controversial. A few authors advocate the use of varnish over glass-ionomer cement with the rationale that the polymerization shrinkage of composite will not

disturb the underlying cement; however, it may affect bonding at the interphase.

Bonding of Composite to Porcelain

Bonding of porcelain to composite is partly because of mechanical interlocking and partly because of chemical union (porcelain inlays and onlays when cemented to the tooth using a resin-based cement). Mechanical retention is achieved by etching the fitting surface of porcelain with dilute hydrofluoric acid/grit blasted with alumina to increase surface roughness. The optimal time for etching depends upon the etchant concentration as well as the type of porcelain used. To improve upon the mechanical attachment, gap between the bonding medium and the porcelain should be minimized by using an intermediate low viscosity resin, which penetrates into the pores by capillary action.

Shear bond strength increases several times when porcelain is etched as compared to when it is not etched.

Chemical union between porcelain and composite is improved by treating the etched surface with a silane coupling agent. However, relatively low increase in bond strength has been observed with this additional treatment suggesting that mechanical interlocking is probably more important than chemical adhesion. Silane coupling agents have also shown to reduce the gap between porcelain and composite, presumably by improving upon the wetting ability; hence, promoting mechanical interlocking. Since the shelf life and durability of silane in moist conditions is questionable, it is preferably used as an aid to mechanical retention.

Bonding of Amalgam to Resin

Bonding of amalgam to resin, amalgam to tooth structure and amalgam to metal has been successfully tried.

Bonded silver restorations offer following advantages:

- Increases retention without the use of additional retentive features, like pins, grooves, dovetails, etc.
- Increases fracture resistance, while conserving tooth structure
- Reduces microleakage
- Decreases chances of recurrent caries

Agents that can be used for bonding amalgam to the substrate are All Bond, Liner Bond-2, Amalgam Bond and Panavia. A specific feature desired in these materials is that they should have dual characteristics to achieve optimal wetting. Dental amalgam is strongly hydrophobic; whereas, enamel is hydrophilic, hence, a wetting agent needs to be incorporated into the bonding resin that can wet both hydrophobic and hydrophilic surfaces. 4-META and MDP are commonly used for this purpose.

The nature of bond between resin and amalgam is mainly micromechanical, as amalgam interlocks with the fluid resin during condensation. Bond strength of amalgam to dentin is relatively low (2–6 MPa). Failure generally occurs at the interface between bonding agent and amalgam; while the bond on the other side, i.e. between resin and tooth is sufficiently strong. The durability and superiority of bonded amalgam over conventional amalgam restorations has not been established.

ADDITIVES FOR BONDING AGENTS

The filler contents, though in minute quantity, play a vital role in dental adhesives. All types of fillers are being used in adhesion. The addition of fillers in total etch adhesives improves bond strength. The self-etching bonding systems contain only small amount of fillers. In single step systems, silica fillers are often used as thickener to increase the viscosity, resulting in adequate film thickness. It also prevents over thinning and incomplete polymerization due to oxygen inhibition.

A few adhesives contain fillers having properties of fluoride release and opacity. Polysiloxane encapsulated sodium fluoride particles are used as a source for fluoride ions. Fluoride is incorporated because of anticariogenic activity, resulting in an increased resistance of enamel and dentin to acid attacks and inhibits the carbohydrate metabolism of dental plaque. The addition of fluoride improves the bond strength.

Certain adhesives use dyes, which indicate proper mixing of the components. In one-up bond, the color changes from yellow to pink. When the adhesive is light cured, the color fades. The addition of antibacterial agents in adhesives is showing promising results. The total etch adhesive system use phosphoric acid to condition the prepared tooth surface by removing the smear layer. The phosphoric acid also exhibits antimicrobial activity, though for a shorter time. Glutaraldehyde has been used as disinfectant in earlier multi-bottle systems, such as Syntac or Gluma-bond as well in i-Bond (one step self-etching system). Another product MDPB (methacryl-oxydodecyl pyridinium bromide) is also used as antibacterial component in various adhesives.

Recently, Nanogel polymer added to resin (BisGMA/HEMA) have shown promising results in improving long-term microtensile bond strength (μTBS). The potential of water-induced phase separation is minimized in the nanogel modified resin.

ROLE OF WATER IN THE BONDING PROCESS

After acid etching, the smear layer is removed and the underlying 3–5 μm layer of dentin is demineralized. Post-etching rinsing with water removes the dissolved minerals and cover a demineralized dentin surface with water. This water maintains the collagen in an expanded state and thereby preserve the spaces needed for infiltration of resin. (Water acts as a plasticizer for collagen and keeps it in a soft state.) If the dentin is exposed to air, water evaporates leaving collagen in a collapsed (stiffened) state; subsequently reducing the ability of the bonding agents to penetrate the collagen fibrils. When the collagen fibrils are brought closer, the secondary forces become active, which is not possible when water is present, thereby increasing the stiffness of collagen. The demineralization results in shrinkage. Upon addition of water, the collagen network re-expands to its original volume. The critical water concentration prevents collapse of the network or allows re-expansion of the dried dentin over a period of 10–30 seconds. When non-aqueous primers are applied over dry dentin, the collagen is not rewetted by water; whereas, when water-based primers are applied to dry dentin, there can be two possibilities: (i) If the water concentration of the primer is low, the resin monomer and/or organic solvent will stiffen the collagen faster than the water can plasticize the collagen; (ii) If the water content of primer is good to plasticize the collagen faster than the resin/solvent stiffen it, the resin monomers will infiltrate the network. Excessive water in the primer should be avoided as it dilutes the monomer concentration. In dry bonding technique, the water-free acetone-based primers do not effectively infiltrate the exposed collagen and may form 'hybridoid regions'. When the dentin is kept moist, the collagen network is maintained in an expanded state and the inter-fibrillar spaces are left open. Bonding to this moist surface could be effective, only if water within the dentin is completely eliminated and replaced by monomers. When applied on wet dentin, the water diffuses into the organic solvent, and penetrates into the demineralized dentin matrix along with the polymerizable monomer. The water is gradually lost as the solvent and resin monomer move further along the collagen fibrils. If the water inside the collagen network is not completely displaced, polymerization of resin inside the hybrid layer may be affected; the remaining water may compete for space with resin inside the demineralized dentin. In such conditions, excess moisture decreases the concentration of organic solvents in the primer (monomer). The presence of monomer over the water layer is more common in the dentinal tubules (high content of water) than in the intertubular dentin ('blister like' structures with water being trapped beneath the resin layer). It is established that a critical amount of water is mandatory for good bonding; however, an overwet condition decreases bond strength by lack of resin tags and formation of blister-like structures at the interface.

The primer should be either water-based or agent miscible in water. The commonly used solvents are:

- Acetone (Prime and Bond NT, Bisco One step)
- Ethanol (OptiBond Solo, 3M Single Bond)
- Water (Scotchbond Multipurpose Plus)

Acetone has a relatively high vapor pressure (184 mm Hg at 20°C) as compared to ethanol (43.9 mm Hg at 20°C) and water (17.5 mm Hg at 20°C). As the solvent evaporates, the viscosity of the dentin bonding agent increases, which decreases the ability of the bonding system to penetrate along the collagen fibers and the dentinal tubules; subsequently inhibiting the proper bond formation.

'Water Treeing' Phenomenon

The adhesives normally contain hydrophilic and/or ionic monomers, which improve their bonding to wet substrate of dentin. It is established that adhesives act as semipermeable membrane, permitting water movement across the polymerized adhesives. It implies that there are interconnected channels within the adhesives, which are responsible for such movement. Usually, these channels are not visible unless incorporated with silver nitrate, since the water is lost after desiccation of the specimens. These water channels (revealed by silver staining), seen at resin–dentin interface, within the adhesive layer, are termed as '*water trees*'. Water trees are generally located along the surface of the hybrid layer extending into the underlying adhesive layers (Fig. 18.28).

Tay and Pashley (2003) hypothesized that water trees within the hybrid layer facilitate leaching of hydrophilic resin components, leading to degradation of resin–dentin bonds.

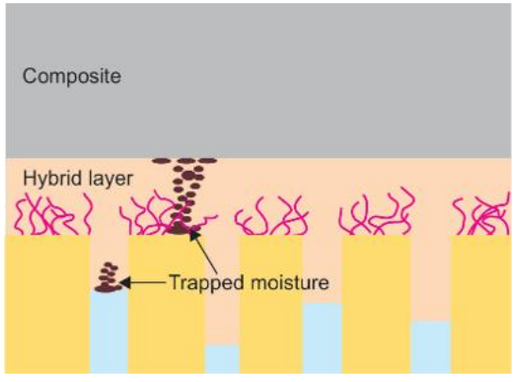


Fig. 18.28: Water tree formation

Two theories have been proposed as regards genesis of water treeing phenomenon.

a. *Remnant Water Theory*

Earlier authors believed that water trees were actually 'left out' water that was incompletely removed from dentin adhesives. Water trees were observed from resin–dentin interface bonded with ethanol-based adhesives and rarely in acetone-based adhesives. It was further observed that addition of HEMA to water lowered the rate of evaporation of water from the HEMA-water mixture.

However, this theory does not account for following aspects of water tree formation.

i. ***Uneven distribution:*** Water trees are never uniformly distributed; seldom seen in the central part of the adhesive layer, rather predominantly present at the surface of the hybrid layer.

ii. ***Substrate dependence:*** Water trees have been frequently observed in sound dentin bonded with one-step self-etch adhesives; however, they were virtually absent when composite resins were used as bonding substrate. It implies that water tree contribute minimally by residual water derived from the dentin adhesive.

iii. ***Reverse water trees:*** Different shapes and forms of water trees have been observed. Usually water-trees originate from the surface of dentin, with their branching pointing upwards into the adhesive. In case of reverse water trees, they appear to originate from water trapped at the interface between the adhesive and the overlying resin composite; spread downwards with the 'branches' pointing towards dentin.

iv. ***Secondary water trees:*** Water droplets are entrapped and seen as clear holes within the adhesive layer. When observed under silver nitrate tracer, water trees were seen radiating circumferentially from the periphery of these water droplets creating a 'sunburst' effect around the droplets.

As water trees seldom exist when water-containing adhesives are used to bond composite-to-composite, it is unlikely that water tree formation is caused by water, that is present within the adhesive formulations. Two features, which could not be explained by this theory, are:

- From where the water is derived?
- What triggers the movement of this additional source of water?

These two features are explained in the 'water-flux' theory.

b. *Water Flux Theory*

The fluid movement in dentin can be of three types, viz. evaporative, osmotic and convective water fluxes. Evaporative is due to air drying. Touch of paper point induces capillary forces leading to outward movement of fluid. Non-vital dentin also contains water; evaporative water flux may occur irrespective of tooth vitality.

The high concentration of water-soluble ionic monomer in the presence of water may also induce osmotic water flux from deep dentin. This may occur just prior to polymerization. As soon as the monomers are converted to polymers, the osmotically induced water flux is ceased.

Both osmotic and evaporative fluxes result in the permeation of water within the adhesive mixture. This permeation explain the presence of normally oriented water trees along the surface of the dentin. It has been shown that following the detection of outward evaporative water flux (*reverse water tree*), after air-drying of the adhesive-coated dentin, additional inward fluid flux would be induced during light activation of the adhesive. During light activation process, some of the water that has reached top of the adhesive layer cannot escape, as it is trapped by the initially polymerized surface of the adhesive. The heat generated by the curing lights reflect water back into the less polymerized subsurface, forming water channels, referred to as '*reverse water tree*'. Water droplets trapped between the adhesive and the resin composite reduce the bonded surface area and increase interfacial stresses; subsequently, dislodge the resin composite (Fig. 18.29).

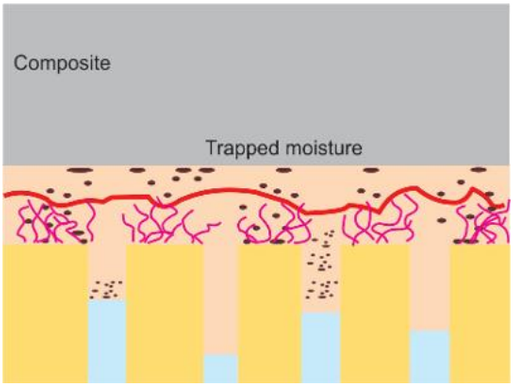


Fig. 18.29: Failure of resin–dentin bond by hydrolytic degradation with passage of time

Transparent Carious Dentin: Absence of Nanoleakage and Water Treeing

Transparent carious dentin is heavily occluded with intratubular mineral deposits. These deposits account for relative impermeability of this layer. The hybrid

layer of transparent dentin is usually three times as thick as that of sound dentin. No nanoleakage can be evident because both evaporative and convective water fluxes from dentin were blocked by heavily occluded dentinal tubules. This may result in excellent initial resin seal (complete absence of water-rich interfacial zones above the bonding substrate, depleted of its mineral contents). It is established that both water treeing and nanoleakage are absent in hybrid layer of transparent dentin. Although the hybrid layer is thicker, it is not possible for any adhesive to completely diffuse through a zone of partially demineralized dentin that may be several hundred micrometers thick.

Nanoleakage and Water Treeing

The inward diffusion of acidic monomer demineralizes the dentin and dissolves the smear layer. The outward osmotically induced water flux may flush out the partially neutralized resin monomer from the demineralized dentin. The presence of such water in acidic adhesive may retard the polymerization of resin within the interfibrillar spaces. The water usually finds its way through the smear layer and is manifested as water trees within the adhesive matrix. Depending upon the chemical composition of the adhesive phase, water may get separated in the form of water droplets, which is manifested as secondary water trees. The water trees are formed along the surface of the hybrid layer. In case of nanoleakage, the subsurface water moves preferentially to pre-existing channels within the hybrid layer.

The reduction in nanoleakage and water tree formation may be achieved when multiple coats of one-step self-etch adhesives are applied on sound dentin. When one-step self-etch adhesives are applied to sound dentin, water droplets may be deposited between the adhesive–composite interface due to rapid water movement across the adhesive. If sufficient layer of adhesive is present, the phenomenon of water blistering may be of minor implications. The adhesives can be rendered less permeable by treating one-step self-etch adhesive as a primer and covering it with a coat of less hydrophilic resin. The coating converts one step into two-step self-etch adhesive and renders them less permeable to water movement.

Acid–Base Resistant Zone at Adhesive–Substrate Interface

It is established that an increased resistance of the tooth–resin interface to acid may effectively retard the ingress of bacterial byproducts, subsequently progression of secondary caries.

An acid–base resistant zone has been reported beneath the hybrid layer in self-etch adhesive systems. This zone can resist acid and base challenges, and play an important role in the prevention of secondary caries. It has been shown that acid–base resident zone is formed in self-etch adhesive systems, and not in etch-and-rinse adhesive systems.

It is hypothesized that penetration of the monomer beyond the hybrid layer coupled with chemical interaction between the monomer and hydroxyapatite may contribute to the formation of acid–base resistant zone.

It has been shown that monomers in self-etch adhesive can chemically interact with the hydroxy-apatite in the demineralized dentin.

Enamel has a higher mineral content as compared to dentin collagen network. The acid–base resistant zone in enamel is not below the hybrid layer, but it is at the interface, which is not dissolved after the acid–base challenge.

SUCCESS/FAILURE OF ADHESIVES

Several variables can affect the clinical performance of adhesive systems vis-à-vis the success/failure of the system (Fig.18.30), such as:

1. *Material factors:* Thorough knowledge of chemistry of the material to be used is mandatory

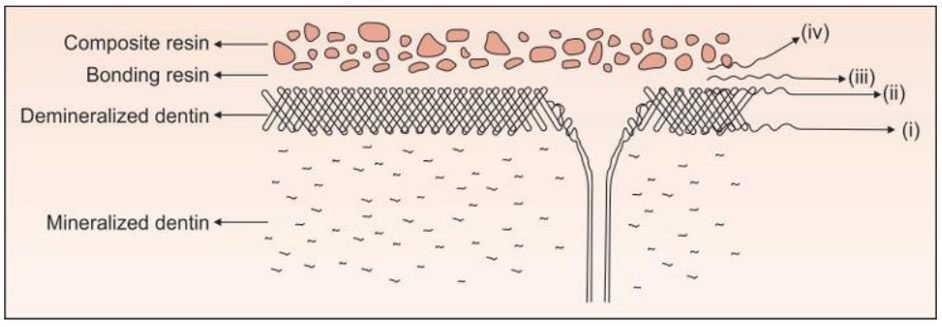


Fig. 18.30: Diagram illustrating different positions of failure at the resin–adhesive–tooth interface: (i) Between the mineralized and demineralized dentin; (ii) Between demineralized dentin and bonding resin; (iii) Within the bonding resin; (iv) Between the bonding resin and composite resin

for successful bonding. Hydrophobic bonding agents do not provide sufficient bond. The manufacturer's instructions should be carefully followed regarding washing off the conditioner and mode of applying the primer and bonding.

2. ***Substrate:*** Variability of the substrate has shown to have a major effect on the clinical performance of adhesive systems. Its details have already been covered in the preceding pages.
3. ***Size and shape of lesion:*** Size of the lesion or the area of dentinal substrate available for bonding is critical for adhesion. Less adhesion is seen in small-sized cervical lesions. Deep wedge-shaped lesions have shown better retention than shallow saucer-shaped lesions.
4. ***Maxillary vs mandibular arch:*** Better adhesion results are expected in the maxillary arch because of lesser chances of moisture contamination; also lower tooth flexure effect in the upper jaws.
5. ***Patient's age:*** With age, the dentin becomes sclerosed (sclerosis leads to decrease in clinical adhesiveness); hence, a higher failure rate. Besides sclerosis, increased tooth flexure may also be a factor towards failure of adhesion. No direct relation between retention failure and patient's age could be established; as sclerosis is pertinent not only to old age, but is determined by the time period for which the dentin has been exposed to the oral environment.
6. ***Tooth flexure:*** Tooth flexure does influence the retention of adhesive restorations, especially the cervical restorations. Heavy centric and eccentric forces are responsible for generating compressive and tensile forces in the cervical area, which may gradually dislodge and debond the resin restoration. The composites with adequate elastic capacity, like microfilled composites, are preferred in such lesions.
7. ***Dentin wetness:*** The principle of good spreading of monomer on dentin is important for successful bonding. Along with wetting, the permeability of the demineralized dentin to monomer also aids in dentin bonding. The bonding agents with effective wetting capability ensure successful bonding.
8. ***Elastic bonding concept:*** Composite resins shrink during polymerization. In order to protect the tooth–composite interface from debonding during polymerization, the intervening adhesive resin should be sufficiently elastic to absorb the polymerization stresses. This can be achieved by using a relatively thick layer of separately polymerized, unfilled or semifilled bonding resins. Alternatively, the additional use of intermediate glass-ionomer liner under composite reduces the total stiffness of the restoration.

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